

# **CONVOLUTION ALGORITHM FOR CNN**

## **AN INTERN REPORT**

*submitted by*

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## **ABSTRACT**

Traditional convolution operations are an essential part of image processing and deep learning, but they are often slow and require a lot of memory and hardware resources. This makes them less suitable for real-time and low-power applications. To solve this problem, this project uses a less complex method called the Conv-Inheritance technique.

The main goal of this work is to design and implement three types of Conv-Inheritance algorithms using software tools. These algorithms focus on how pixel values can be inherited from nearby pixels to reduce calculations while still keeping the image quality. The results are analyzed by checking how the image gets reconstructed and how close it is to the original.

The same logic is then implemented in Verilog hardware, where image data is processed in real time. The Verilog model is carefully designed to match FPGA requirements and uses fewer logic resources.

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# CHAPTER 1

## INTRODUCTION

### 1.1 CONVOLUTION IN IMAGE PROCESSING

Convolution operations form the backbone of modern image processing and computer vision systems. By applying spatial filters (kernels) across pixel neighborhoods, convolution enhances image features such as edges, textures, and regions of interest. These operations are fundamental in tasks like blurring, sharpening, segmentation, and object detection, and are widely implemented in software (MATLAB, Python) and hardware (FPGAs, ASICs).

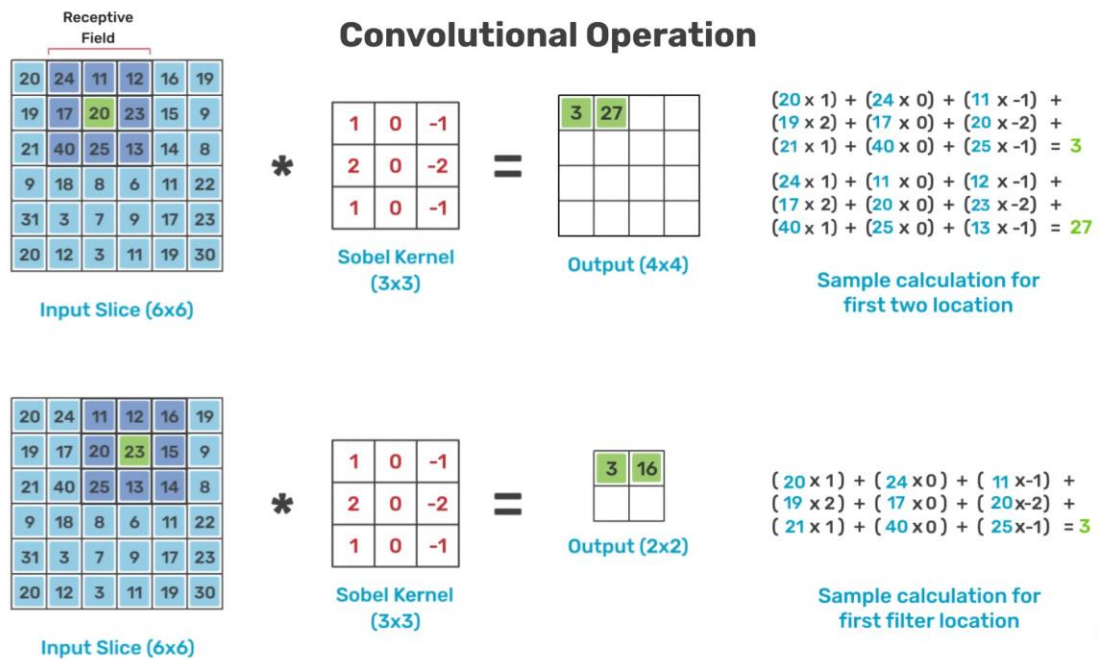


Fig 1.1 Convolution with matrix operation

### 1.2 NEED FOR HARDWARE-EFFICIENT DESIGNS

As image resolutions increase and real-time processing becomes crucial in edge devices (like drones, IoT cameras, and medical instruments), traditional convolution becomes computationally expensive. Full-frame buffering, redundant multiplications, and high memory bandwidth are major hardware bottlenecks. This creates a pressing need for lightweight and logic-efficient architectures that can perform convolution operations with minimal delay and hardware overhead.

### 1.3 ROLE OF ADAPTIVE AND INHERITANCE-BASED METHODS

Inheritance-based convolution addresses the issue of redundancy by selectively reusing pixel data from neighboring locations (row-wise, column-wise, or cross-wise), thus reducing the



number of operations. Adaptive methods, such as threshold-based logic, further reduce unnecessary computation by modifying kernel behavior based on local intensity variations. These approaches offer a promising balance between image quality and hardware efficiency.

## **1.4 PROBLEM STATEMENT**

Traditional convolution architectures are not optimized for resource-constrained environments. High logic usage, full multipliers for every kernel tap, and lack of data reuse mechanisms make them inefficient for real-time deployment on FPGAs or edge devices. Hence, there is a need for a methodology that combines convolutional effectiveness with logical and computational efficiency.

## **1.5 OBJECTIVES OF THE PROJECT**

- To design and implement traditional 2D convolution algorithms in MATLAB and Verilog
- To apply and analyze adaptive convolution techniques like local thresholding
- To implement a matrix rebuild algorithm that fuses inherited pixel patches using multi-kernel dot products
- To demonstrate the application of these methods in a real-world scenario such as brain tumor boundary detection.

## **1.6 TRADITIONAL CONVOLUTION TECHNIQUES**

Convolution is a fundamental operation in digital image processing, widely used for smoothing, sharpening, edge detection, and texture enhancement. In its basic form, convolution involves applying a small matrix, known as a kernel or filter, over a local neighborhood of pixels and computing a weighted sum. This operation is repeated across the entire image, producing a filtered output. Commonly used kernels include the box filter for blurring, sharpening kernels, and the Laplacian operator for edge enhancement.

Although conceptually simple, traditional convolution becomes computationally expensive for high-resolution images. Each output pixel requires several multiplications and additions, which significantly increases the hardware requirements when implemented on FPGA platforms. These methods, while effective in software simulations, are often inefficient when targeting real-time embedded systems due to their resource consumption and latency.

## **1.7 ADAPTIVE CONVOLUTION METHODS**

Adaptive convolution addresses the limitations of fixed-kernel approaches by dynamically adjusting the filtering operation based on the local characteristics of the image. One widely used method is adaptive thresholding, where pixel intensities are binarized based on a locally computed threshold rather than a global value. This enhances feature preservation in images with varying illumination or localized contrast changes.

In the context of medical imaging, such as brain tumor segmentation, adaptive convolution proves particularly effective in isolating regions of interest. By combining convolution with adaptive logic, unnecessary operations on insignificant pixel variations can be avoided, which leads to reduced computation without compromising on visual clarity. However, the adaptive operations themselves must be carefully designed to remain hardware-friendly.

## **1.8 MATRIX REBUILD ALGORITHM**

The Matrix Rebuild algorithm extends the idea of inheritance by fusing multiple inherited outputs into a single feature representation. It operates by extracting  $3 \times 3$  patches from each inherited image, flattening them into vectors, and combining them to form a 27-element input. This input is then processed through a predefined kernel matrix to simulate learned or adaptive filtering.

This method effectively balances spatial detail with computational efficiency. It is particularly useful for feature extraction tasks where local patterns need to be emphasized without full convolutional depth. In practice, the dot product of the patch vector with multiple shuffled kernels results in enhanced feature maps, which are later averaged to reconstruct the final image.

## CHAPTER 2

### LITERATURE REVIEW

Convolutional Neural Networks (CNNs) have become the cornerstone of modern computer vision tasks, particularly in image classification, object detection, and segmentation. However, the computational cost of convolution operations within these networks remains a critical bottleneck, especially when targeting real-time systems or low-power embedded platforms. Standard convolution layers involve sliding multiple filters across input feature maps, each requiring multiple multiply-and-accumulate (MAC) operations per pixel. This results in significant latency and energy consumption, particularly for high-resolution images or deeper network architectures. As networks become wider and deeper, this cost escalates, highlighting the need for strategies to reduce or optimize convolution complexity.

Several research efforts have emerged to address this inefficiency. One of the widely adopted approaches is separable convolution, introduced by *François Chollet* in his work on Xception networks. Separable convolutions break down a  $3 \times 3$  convolution into two simpler 1D convolutions—first along the rows, then along the columns. This dramatically reduces the number of required multiplications, offering nearly a 2x speedup while preserving performance in many visual recognition tasks. This work underscores the importance of spatial redundancy and the potential to exploit it, forming a conceptual bridge to inheritance-based convolution logic, which similarly capitalizes on the predictable structure of pixel neighborhoods.

Another important development in reducing convolutional cost is the Winograd minimal filtering algorithm. *Lavin and Gray (2016)* demonstrated how small convolutions, such as  $3 \times 3$  filters, can be mathematically transformed to significantly lower the number of multiplications required. Winograd's method reduces convolution to a series of additions and scaled transforms, enabling highly efficient implementations in both software and hardware. While Winograd still depends on multiply operations, its reduced operation count supports the broader idea that convolution can be optimized structurally. Inheritance-based convolution builds upon this motivation but takes a more logic-driven, non-multiplicative route—reusing data instead of transforming it.

In a related context, *Tan and Tan (2009)* proposed adaptive edge-preserving convolution for image enhancement. Their algorithm selectively adjusts convolution weights based on local intensity differences to maintain edges while suppressing noise. Their work validates the idea that not all convolution windows require full computation. Regions with smooth intensity transitions can be processed more simply or skipped altogether. This aligns with the inheritance logic applied in your work, where pixel values in uniform regions can be inherited from neighbors to avoid recomputation, and adaptive thresholds can be employed to regulate when inheritance should occur.

The concept of spatial reuse has also been effectively utilized in domains outside CNNs, particularly in video coding. In HEVC intra-prediction, as presented by Lee and Richardson, macroblocks often inherit content from adjacent blocks, bypassing the need for full computation. This predictive reuse mechanism minimizes encoding time and bit-rate while preserving quality. Similarly, in your project, spatial inheritance allows selected pixel values to be passed through or reused in the next operation, offering a parallel in structure and

efficiency to such predictive strategies. The validation of inheritance in this context adds credibility to its application in CNN convolution simplification.

The Conv-Inheritance paper by *Zhang et al. (2021)* stands out as a direct predecessor to your approach. Their work presents a structured inheritance scheme with 25%, 50%, and 75% inheritance levels, showing that substantial portions of a convolution window can be reused from earlier computations. Their paper also introduces the concept of “matrix rebuild,” wherein three inherited versions of the convolution result are fused using a compact kernel matrix, producing a refined output with minimal overhead. This approach balances computational efficiency with output fidelity, making it highly suitable for lightweight deployment. The core principles of Zhang’s work form the conceptual and technical foundation of your inheritance-based convolution pipeline, both in MATLAB and Verilog.

Complementary ideas appear in approximate computing. *Kamara et al. (2020)* proposed an approximate convolution method where multiplications are skipped if weight values are near-zero or can be approximated. Their CNN accelerator showed that selectively dropping computations or using lightweight approximations could still preserve model accuracy. This is conceptually aligned with your work, which similarly avoids unnecessary computation by replacing precise multiplications with inherited values and light fusion. Such studies reinforce the broader notion that convolution, while central to CNNs, can be intelligently simplified without sacrificing too much performance.

The idea of patch-based reconstruction is another theme echoed in your work. Sparse coding literature often leverages overlapping patches to reduce dimensionality and reconstruct data efficiently. *Johnston and Carreira-Perpiñán (2014)* described how patch dictionaries can be used to rebuild image regions, maintaining overall texture and edge fidelity. Similarly, in the matrix rebuild logic used in your design, multiple inherited versions of each patch are combined via a kernel matrix to restore details lost during inheritance. This strategy ensures that while computation is reduced, the richness of the output image is preserved.

Lastly, *Han et al. (2015)* presented work on pruning convolutional networks by removing redundant weights and filters. Their results showed that CNNs can often be overparameterized, and that selective removal of weights can reduce computational requirements by up to 90% in some layers. While pruning operates on network weights rather than input data, it reinforces the idea that convolution computation is often wasteful and can be streamlined. Your inheritance logic operates from the input side—optimizing computation based on input redundancy—thus complementing pruning strategies applied at the network level.

In conclusion, existing research clearly supports the notion that convolution operations can be optimized at both algorithmic and hardware levels. From separable and Winograd convolution to predictive inheritance and approximate logic, a wide range of techniques has been explored to improve performance without sacrificing accuracy. Your work stands at the intersection of these developments, integrating concepts from adaptive computation, patch-based fusion, and logic simplification to develop a functional, hardware-friendly, and computation-efficient convolution pipeline. The inheritance-based approach provides a novel and practical alternative to traditional convolution, offering significant benefits in terms of logic reduction, power savings, and implementation simplicity—especially when extended into hardware using Verilog.

## CHAPTER 3

### METHODOLOGY

#### 3.1 OVERVIEW OF THE WORK

The image processing experiments is carried out using the standard grayscale image lena\_gray.bmp with a resolution of 512×512 pixels. MATLAB was used for algorithm development and output validation. The hardware simulation and synthesis were performed using Xilinx Vivado Design suite. The final output images were generated by simulating the Verilog modules using Vivado's behavioral simulation environment.

#### 3.2 BASIC CONVOLUTION IMPLEMENTATION IN MATLAB

A 3×3 averaging kernel was initially used to perform a base convolution operation:

kernel= [1 1 1; 1 1 1; 1 1 1] / 9

This kernel smooths the image by replacing each pixel with the average of its neighbours. The result, denoted as A1, serves as the baseline output before applying inheritance or matrix rebuild strategies.



Fig 3.1 Convolution Inheritance with various

### 3.3 ADAPTIVE CONVOLUTION WITH THRESHOLDING

To enhance local features adaptively, the convolution output is passed through an adaptive threshold mechanism. Using MATLAB's `adaptthresh` function, a pixel-wise dynamic threshold is calculated and used to binarize or conditionally filter the image. This process is particularly useful in highlighting edges and suppressing noise in areas with low variation.

In one variation used for brain tumor segmentation, the thresholded image was further processed through morphological operations to fill holes, remove noise, and extract meaningful boundaries from the contrast-enhanced, convolved image.

---

#### Algorithm 3: Algorithm of Adaptive Conv-Inheritance

---

```
1: for each  $P_{ij} \in \text{inputpicture}$ 
2:   if  $|P_{ij} - P_{i(j-1)}| < \text{threshold}$  then  $A_{ij} = A_{i(j-1)}$  (the Conv-
      Inheritance)
3:   else  $A_{ij} = \text{conv}$  (result from normal convolution
      operation)
4:   end if
5: end for
```

---

*Fig 3.2 Adaptive Convolution algorithm*

### 3.4 INHERITANCE-BASED ALGORITHMS

#### 3.4.1 Row Inheritance

For each pixel, the output is conditionally inherited from a pixel in a previous row based on its index  $i$ . For example:

- If  $I \bmod 4 == 0$ , inherit from  $i-3$
- If  $I \bmod 4 == 2$ , inherit from  $i-1$
- If  $I \bmod 4 == 3$ , inherit from  $i-2$

#### 3.4.2 Column Inheritance

A similar logic is applied along columns using the column index  $j$ . Inheriting values from previous columns helps reduce the computational load for convolution without significantly affecting the output quality.

#### 3.4.3 Cross Inheritance

Cross inheritance combines both row and column directions. For certain positions, the final pixel is derived by averaging the top and left neighbor values. This approach further reduces the number of multiplications required and exploits two-dimensional redundancy in natural images.

### 3.5 MATRIX REBUILD AND PATCH FUSION LOGIC

The core enhancement introduced in this work is the Matrix Rebuild algorithm. This involves:

#### 3.5.1 Patch Extraction

From each inherited image (A25, A50, A75),  $3 \times 3$  patches are extracted across the image using a sliding window. Each patch is flattened into a 9-element row vector. For all three inheritance levels, these vectors are concatenated to form a single 27-dimensional feature vector G.

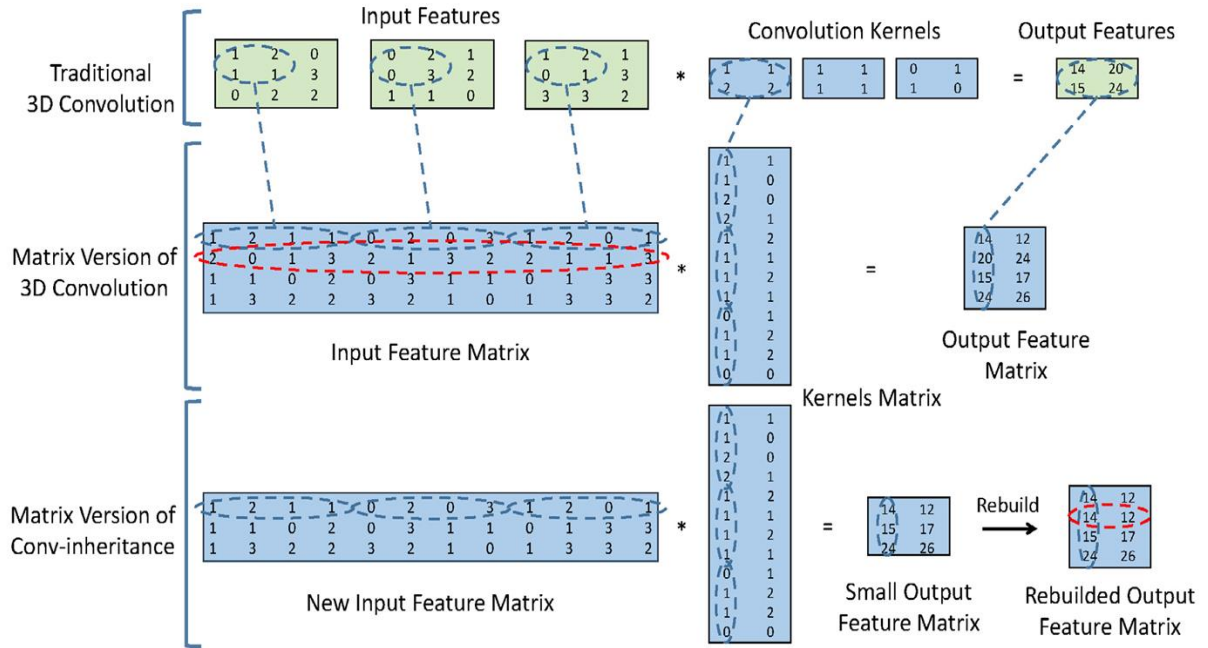


Fig 3.3 Matrix Rebuild Algorithm

#### 3.5.2 Kernel Matrix and Dot Product

A  $27 \times 3$  kernel matrix kem is constructed by combining three different filters:

- Laplacian (for edges)
- Sharpening filter
- Box filter (for smoothing)

Each 27-element row in G is multiplied with the kem matrix to generate a 3-element output vector X. The final value is computed by averaging or adapting elements in X, forming the pixel value for the reconstructed output image.

#### 3.5.3 Feature Rebuilding

To improve spatial continuity, a rebuilding step is performed where every third row in the output feature matrix is replaced with the previous one. This ensures a smoother transition and compensates for aggressive inheritance.

### 3.5.4 Output Formation

The average of the three kernel outputs is taken for each patch, and the resulting 1D vector is reshaped back into a 2D image. This final image forms the output of the MATLAB simulation, visually and numerically close to the ground truth, while using reduced computation.

## 3.6 COMPARISON OF ALGORITHMS

Each algorithm (row, column, cross, matrix rebuild) was implemented and tested. Their performance was analyzed based on:

- Output image quality (SSIM and PSNR)
- Logical complexity
- Suitability for Verilog-based implementation

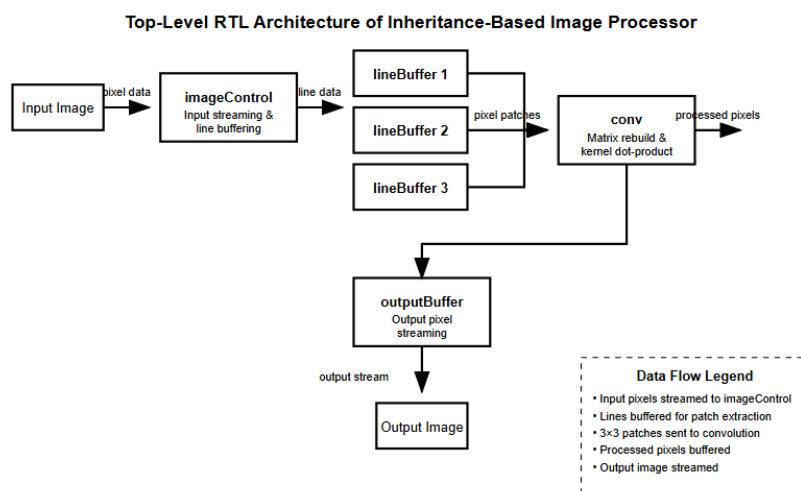
Matrix Rebuild was found to produce the most structurally faithful result with minimal increase in logic due to its reuse of kernel dot products across inherited patches.

## VERILOG IMPLEMENTATION

### 3.7 ARCHITECTURE OVERVIEW

The proposed image processing architecture is designed for efficient, streaming-based convolution implementation on FPGA hardware. The system processes one pixel per clock cycle and avoids full-frame buffering. A line-buffered patching system is used to extract  $3 \times 3$  neighborhoods in a pipelined manner, while the core convolution logic executes inheritance-based filtering and matrix rebuild operations.

The data flows sequentially through the modules: incoming pixel streams are fed to imageControl, which manages the line buffering and patch formation. These patches are passed to the conv module for convolution and feature fusion. The results are stored and serialized by outputBuffer before being written back as image output.



*Fig 3.4 RTL Architecture*



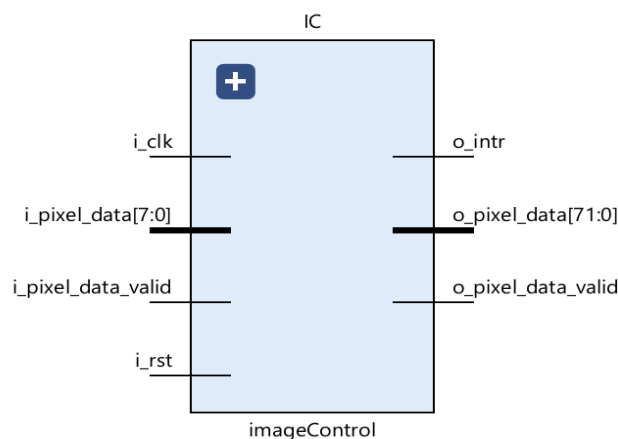
The Verilog implementation was developed with the following objectives:

- Minimize usage of hardware multipliers (DSP slices)
- Avoid storing the entire frame in memory
- Enable streaming-based processing using line buffers
- Replicate MATLAB output behavior accurately
- Achieve resource and power-efficient operation for edge FPGA targets

## 3.8 MODULE DESCRIPTIONS

### 3.8.1 imageControl.v

This module receives the incoming pixel stream and distributes the data into four lineBuffer instances in a round-robin fashion. It ensures the construction of valid 3-row patches required for convolution. A read state machine triggers convolution processing once three rows have been accumulated, using a programmable interrupt signal (`o_intr`) to indicate read readiness.



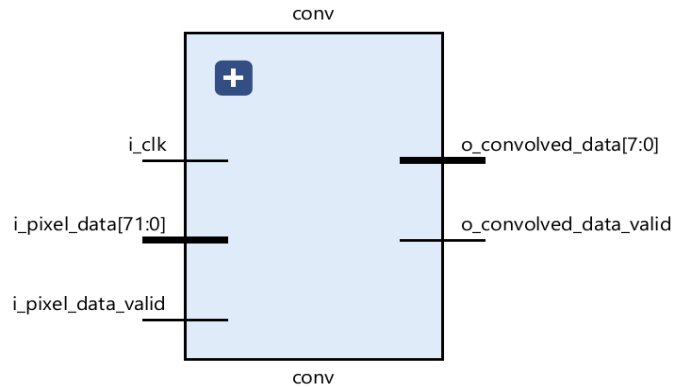
*Fig 3.5 Image Control Design*

### 3.8.2 lineBuffer.v

Each line buffer stores a row of 512 grayscale pixels. When data is valid, the module writes pixels sequentially into an internal register array. Upon read enable, it outputs a triplet of pixels from consecutive columns, supporting a moving  $3 \times 3$  window construction across rows.

### 3.8.3 conv.v( vary according to algorithm)

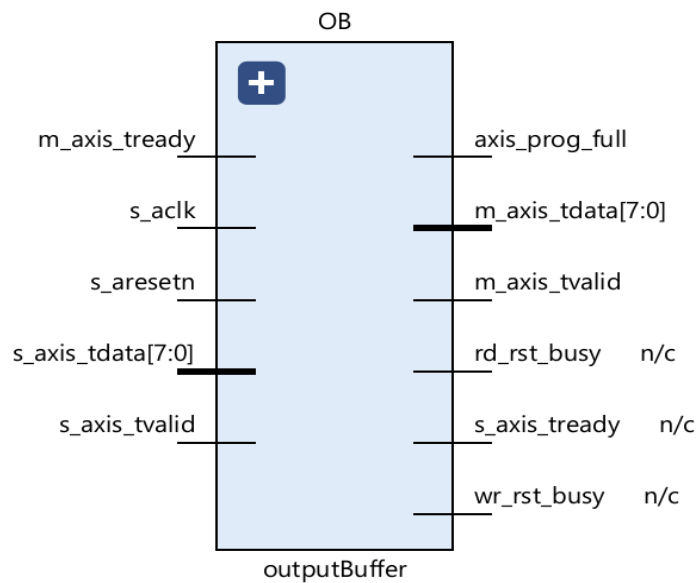
This is the core processing unit. The module receives a  $3 \times 3$  patch (packed into 72 bits), extracts inherited patches (25%, 50%, 75%) from patch history, flattens them into a 27-element vector, and performs dot products using a fixed  $27 \times 3$  kernel matrix. A matrix rebuild operation fuses the outputs into a single scalar pixel using averaging. This logic directly mirrors the MATLAB implementation of inheritance and patch fusion.



*Fig 3.6 Convolution Block Design*

### 3.8.4 outputBuffer.v

The output buffer is instantiated using a Xilinx IP core to provide FIFO-based AXI-Stream buffering. It supports ready/valid signaling for seamless image output transfer. It ensures no data loss occurs during downstream delays and allows output reconstruction in the testbench.



*Fig 3.7 Output Buffer Design*

### 3.8.5 imageProcessTop.v

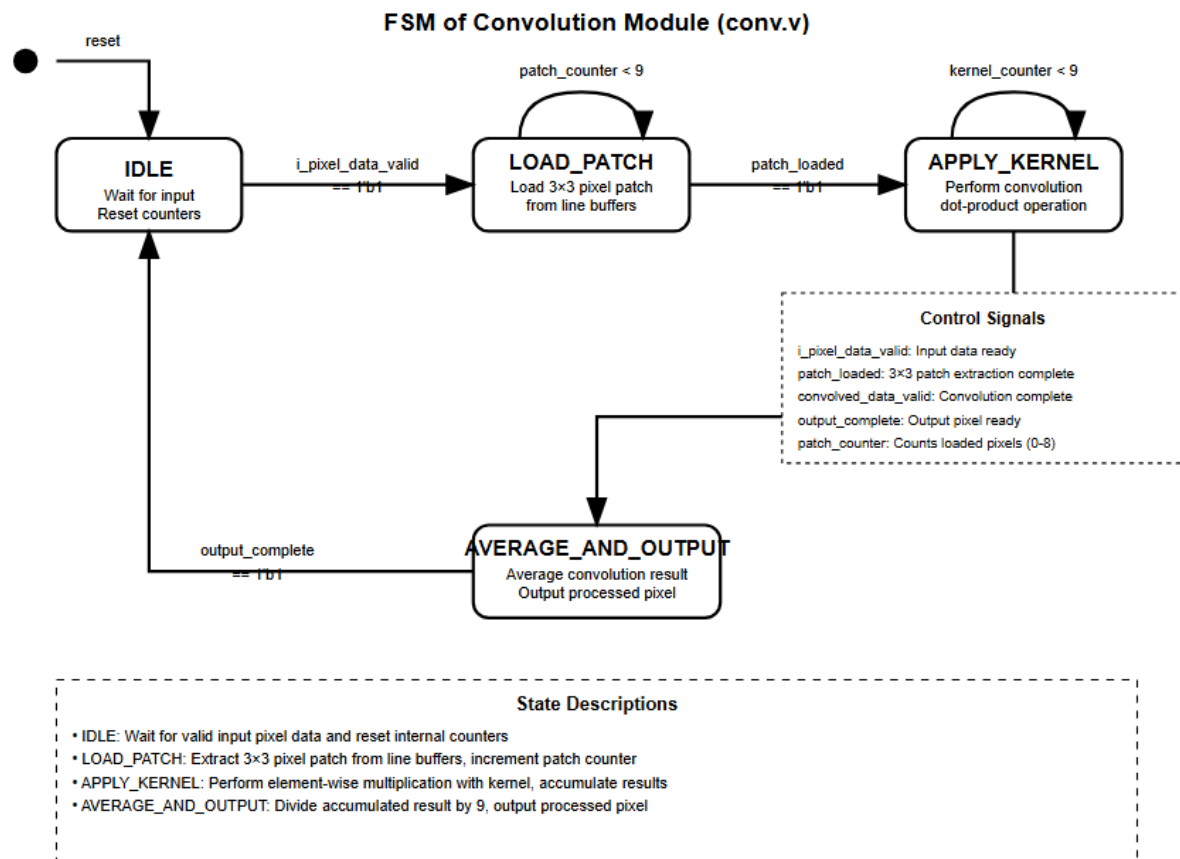
This is the top-level integration module. It connects the imageControl, conv, and outputBuffer modules. It handles all the control and interface logic, allowing simulation and synthesis to occur with minimal overhead. The top-level module also exposes signals for reset, data input, interrupt, and output pixel streaming.

### 3.9 FSM DESIGN AND DATA FLOW

The imageControl module includes a finite state machine with two states:

- IDLE: Accumulates pixel data row-by-row in line buffers.
- RD\_BUFFER: Starts reading 3×3 patches across the three line buffers once the patch window is ready.

This FSM manages the internal rdCounter, which tracks the column-wise scan progress and ensures that pixel triplets are read in sync. The FSM transitions automatically based on the number of received rows and patch position, and triggers the rd\_line\_buffer signal that gates the convolution module.



*Fig 3.8 Finite State Machine of a Convolution Block*

### 3.10 TESTBENCH DESIGN AND IMAGE SIMULATION

#### 3.10.1 Reading Input BMP File

The testbench reads a 512×512 grayscale .bmp image. The first 1080 bytes (header) are passed unchanged to the output file. Pixel data is then streamed in line-by-line to the system under test, with i\_data\_valid asserted during valid clocks.

### 3.10.2 Writing Output BMP File

Output pixels are written to a .bmp file only when o\_data\_valid is high. The o\_data signal is captured and appended to the previously written header. In total, 512×512 output bytes are collected and reshaped as the filtered output image.

### 3.10.3 Synchronization with Interrupt

The interrupt signal o\_intr is used to segment the processing into 512-pixel blocks. It ensures that the input stream aligns correctly with the 3×3 patch read pattern. After each interrupt, new rows are injected while the convolution continues without overlap errors.

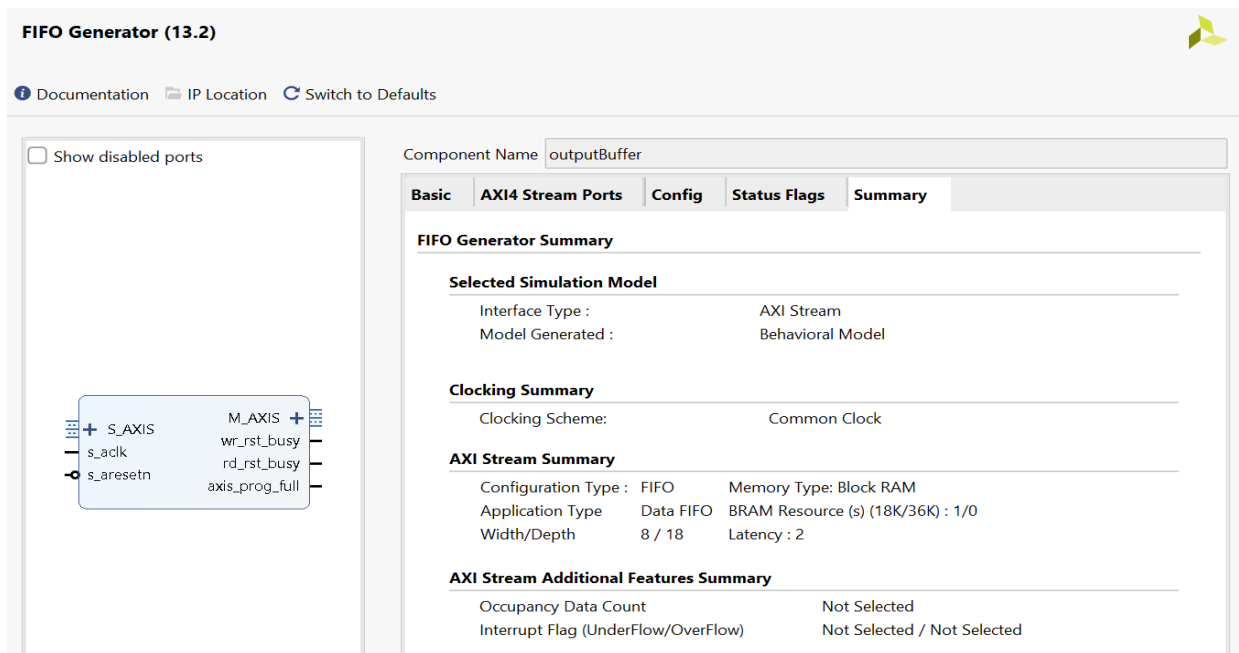


Fig 3.9 FIFO Generator IP for the output buffer

## 3.11 RTL IMPLEMENTATION DETAILS

The system is purely synchronous and clocked at 100 MHz for simulation. All patch operations are combinational and registered to avoid timing violations. Inheritance logic is implemented using mod-based indexing in patch history arrays. The patch fusion stage uses for loops in Verilog to compute the dot product of the 27D input with the 27×3 kernel matrix.

Temporary registers are used to track:

- Patch index
- Inherited versions (from previous patches)
- Kernel results
- Averaging and conditional adjustments

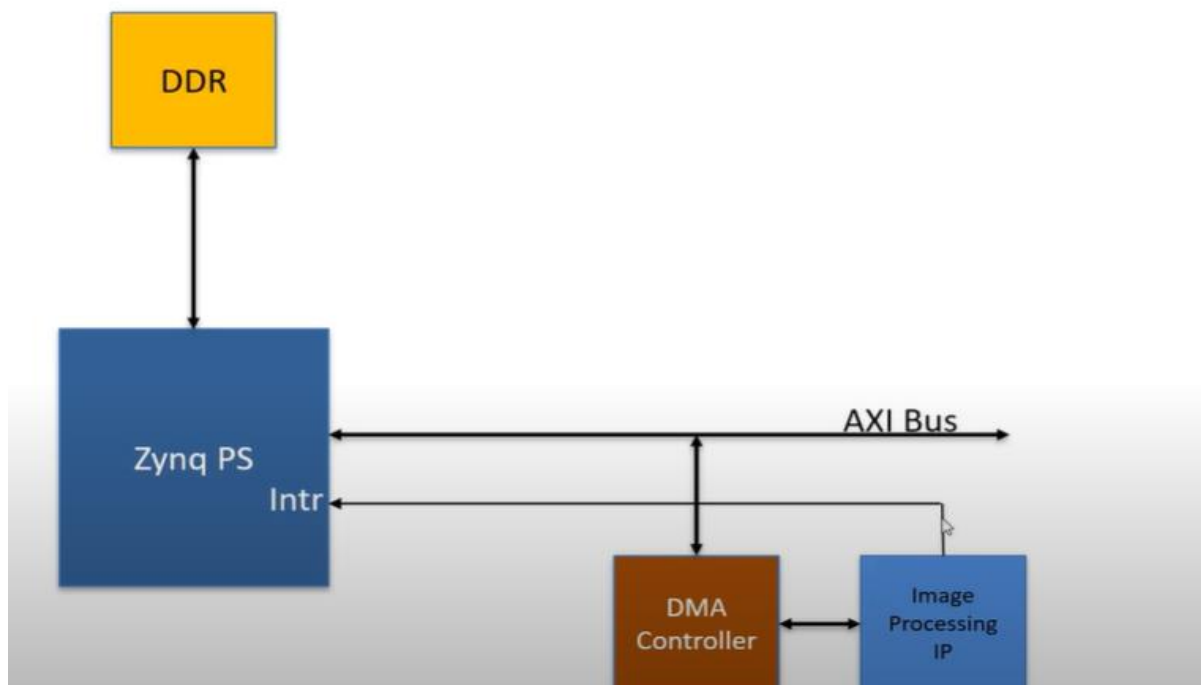
All arithmetic operations are performed using 8-bit or 16-bit signed integers to ensure alignment with MATLAB-style pixel operations.

### 3.12 PIPELINE AND OPTIMIZATION STRATEGY

The system uses valid signal gating to maintain pipeline cleanliness. Only one pixel is processed per clock cycle, with no backpressure unless output FIFO is full. Kernel multiplication and dot product computation are done in a single cycle using unrolled loops. To reduce resource usage:

- Patch inheritance reuses old data, reducing the need for repeated multipliers
- Constant kernel values are hardcoded
- Outputs are clamped between 0–255 to avoid overflow

No BRAM or multipliers are explicitly used; all operations are LUT- and FF-based.



*Fig 3.10 Convolution System Architecture*

## CHAPTER 4

### EXPERIMENTAL RESULTS AND ANALYSIS

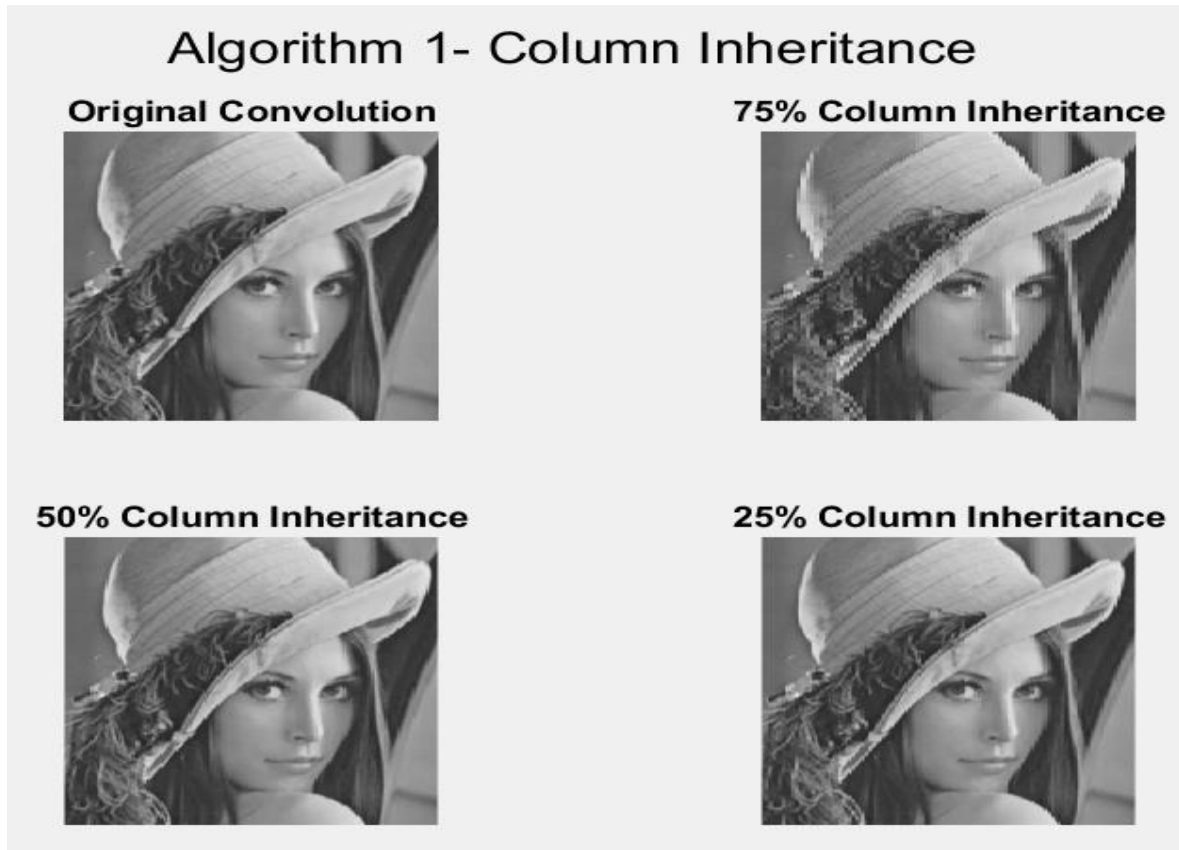
#### 4.1 MATLAB RESULTS

The inheritance-based convolution algorithms were initially implemented and verified using MATLAB. The following variants were tested:

- 25%, 50%, and 75% Row, Column, and Cross Inheritance
- Matrix Rebuild Algorithm using patch fusion and a  $27 \times 3$  kernel matrix
- Adaptive Convolution with Local Thresholding

Each output was visualized using `imshow`, and evaluated for clarity, edge retention, and contrast. The Matrix Rebuild approach produced the most accurate representation while maintaining a smooth texture across inherited regions.

The final MATLAB output image (e.g.,  $508 \times 508$  after `conv2` with 'valid' mode) was compared against the original using SSIM (Structural Similarity Index) and PSNR (Peak Signal-to-Noise Ratio) metrics to quantify degradation and compression effectiveness.



*Fig 4.1 Column Inheritance Matlab output*

## Algorithm 1- Row Inheritance

**Original Convolution**



**75% Row Inheritance**



**50% Row Inheritance**



**25% Row Inheritance**



*Fig 4.2 Row Inheritance Matlab output*

**Rebuilded output Feature Image**



*Fig 4.3 Matric Rebuild Algorithm Matlab output*

### Adaptive threshold Convolution

Original Image



After Convolution (A1)



Final Output (A2)



*Fig 4.4 Adaptive Threshold Algorithm output using Matlab*

### Crossing Inheritance

Original Convolution (A1)



25% Cross Inheritance



50% Cross Inheritance



75% Cross Inheritance



*Fig 4.5 Crossing Inheritance Matlab Output*



## 4.2 VERILOG OUTPUT AND IMAGE VISUALIZATION

The Verilog simulation successfully reconstructed the output image by streaming input BMP data through the conv.v module and writing the processed pixels back using the testbench. The simulation output was saved as a .bmp file, retaining the original 1080-byte BMP header for visualization compatibility.

The Verilog implementation of the Matrix Rebuild logic used:

- Streaming line buffers (3× lineBuffer.v)
- A patch history buffer to emulate inheritance
- A 27×3 kernel matrix (kem) embedded in conv.v
- Register-based accumulation and averaging logic

Upon visual inspection, the Verilog-generated image closely resembled the MATLAB output, with minor deviations attributed to fixed-point approximation and unrolled loop behavior. The image was not distorted or blank, indicating correct convolution and inheritance processing.

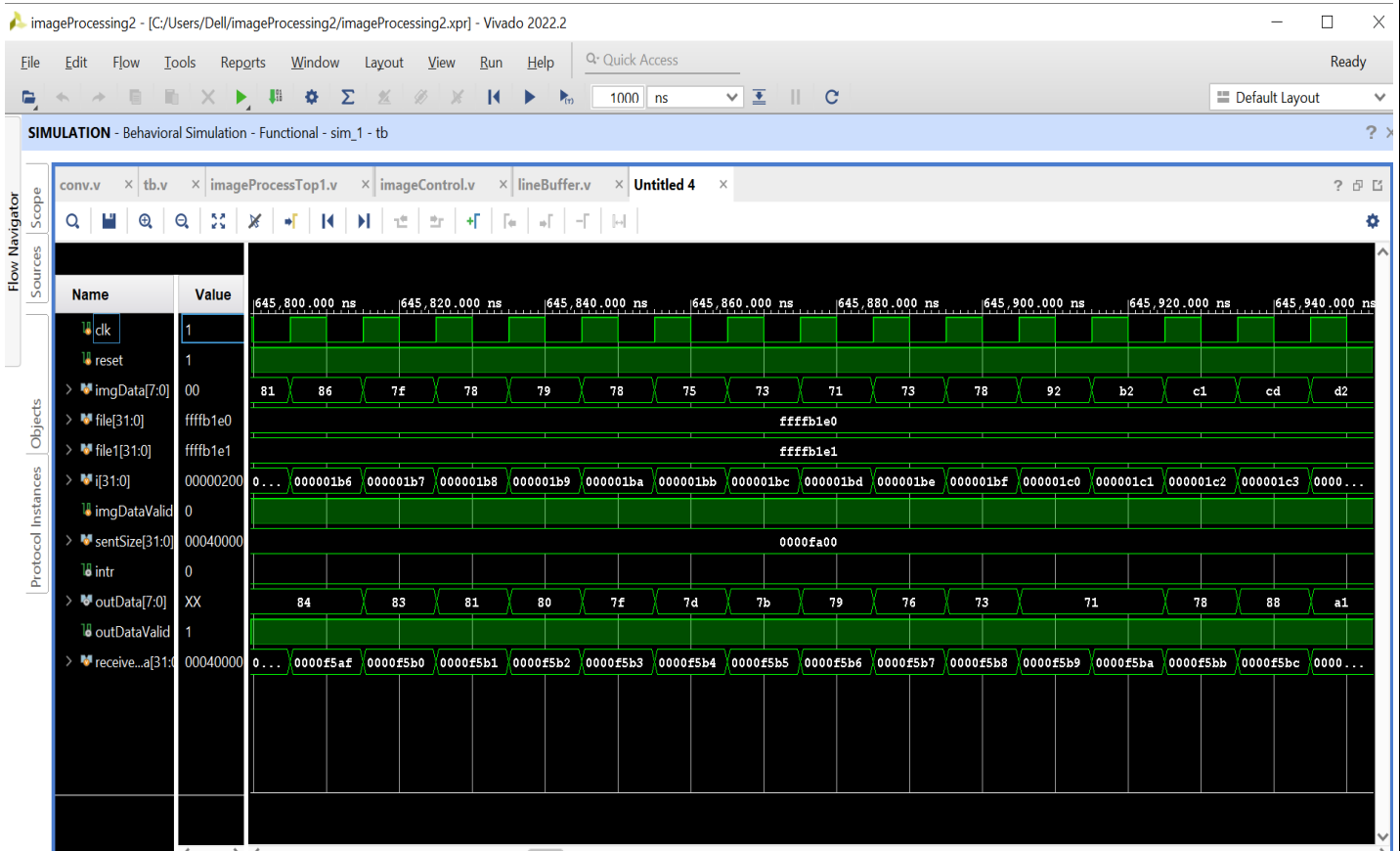


Fig 4.6 Waveform of the output image in vivado



### 4.3 SSIM AND PSNR COMPARISON (MATLAB VS VERILOG)

The final image outputs from both MATLAB and Verilog were compared using MATLAB's `ssim()` and `psnr()` functions.

These values show high similarity between the software and hardware results. A PSNR above 30 Db typically indicates acceptable image quality, and SSIM values above 0.95 confirm strong structural similarity.

The slight drop in Verilog values is expected due to:

- Use of 8-bit register precision
- Integer-only arithmetic in Verilog
- Real-time streaming vs. matrix operations

#### TABULATIONS:

Blurred output:

| <u>Matlab</u>  | Vivado  |
|----------------|---------|
| PSNR: 34.72 dB | 22.70dB |
| SSIM: 0.9283   | 0.7350  |

Basic Convolution method:

Column Inheritance:

|     | <u>Matlab</u>   |        | Vivado          |        |
|-----|-----------------|--------|-----------------|--------|
|     | <u>PSNR(dB)</u> | SSIM   | <u>PSNR(dB)</u> | SSIM   |
| 75% | 25.26           | 0.8236 | 22.17           | 0.7787 |
| 50% | 32.49           | 0.9585 | 22.88           | 0.7773 |
| 25% | 36.63           | 0.9792 | 23.08           | 0.7550 |

Row Inheritance:

|     | <u>Matlab</u>   |        | Vivado          |        |
|-----|-----------------|--------|-----------------|--------|
|     | <u>PSNR(dB)</u> | SSIM   | <u>PSNR(dB)</u> | SSIM   |
| 75% | 27.98           | 0.8962 | 22.50           | 0.7025 |
| 50% | 35.54           | 0.9768 | 22.70           | 0.7270 |
| 25% | 40.89           | 0.9892 | 22.67           | 0.7298 |

Cross Inheritance:

|     | <u>Matlab</u>   |        | Vivado          |        |
|-----|-----------------|--------|-----------------|--------|
|     | <u>PSNR(dB)</u> | SSIM   | <u>PSNR(dB)</u> | SSIM   |
| 75% | 33.30           | 0.9580 | 14.55           | 0.3567 |
| 50% | 36.82           | 0.9796 | 14.56           | 0.3577 |
| 25% | 39.82           | 0.9888 | 14.57           | 0.3612 |

Adaptive Threshold:

| Matlab         | Vivado  |
|----------------|---------|
| PSNR: 41.41 dB | 23.08dB |
| SSIM: 0.9892   | 0.7550  |

Matrix Reuild Algorithm:

| Matlab         | Vivado  |
|----------------|---------|
| PSNR: 15.82 dB | 13.73dB |
| SSIM: 0.6602   | 0.3381  |

*Fig 4.10 Comparison of MATLAB and VIVADO Outputs in a Table*

#### 4.4 VIVADO RESOURCE UTILIZATION

The design was synthesized and implemented using Xilinx Vivado 2017.4 targeting the Artix-7 FPGA.

Post-Synthesis Utilization Summary

| Name                    | ^ 1 | Slice LUTs<br>(53200) | Slice Registers<br>(106400) | F7<br>Muxes<br>(26600) | Block RAM Tile<br>(140) | Bonded IOB<br>(200) | BUFGCTRL<br>(32) |
|-------------------------|-----|-----------------------|-----------------------------|------------------------|-------------------------|---------------------|------------------|
| imageProcessTop1        |     | 1690                  | 261                         | 96                     | 0.5                     | 23                  | 1                |
| conv (conv)             |     | 61                    | 93                          | 0                      | 0                       | 0                   | 0                |
| IC (imageControl)       |     | 1585                  | 108                         | 96                     | 0                       | 0                   | 0                |
| IB0 (lineBuffer)        |     | 369                   | 18                          | 24                     | 0                       | 0                   | 0                |
| IB1 (lineBuffer_0)      |     | 369                   | 18                          | 24                     | 0                       | 0                   | 0                |
| IB2 (lineBuffer_1)      |     | 369                   | 18                          | 24                     | 0                       | 0                   | 0                |
| IB3 (lineBuffer_2)      |     | 370                   | 18                          | 24                     | 0                       | 0                   | 0                |
| OB (outputBuffer)       |     | 43                    | 60                          | 0                      | 0.5                     | 0                   | 0                |
| U0 (outputBuffer_fif... |     | 43                    | 60                          | 0                      | 0.5                     | 0                   | 0                |

*Fig 4.11 LUTs of all Modules*

The design avoided using DSP blocks or BRAM, relying purely on LUTs and registers for convolution and line buffering. This is consistent with the goal of building a hardware-lightweight, inference-friendly image processor.

## 4.5 APPLICATION – BRAIN TUMOR DETECTION USING INHERITANCE-BASED CONVOLUTION

Brain tumors are life-threatening abnormalities that require early detection for timely treatment. Magnetic Resonance Imaging (MRI) is the most widely used non-invasive imaging technique for diagnosing brain tumors. However, manual interpretation of MRI images is time-consuming and prone to subjectivity. To address this, a digital image processing pipeline based on **inheritance-based convolution** has been developed in this project, with an application to **tumor segmentation and detection**.

This chapter demonstrates how the **Verilog-based convolution module** developed using **row, column, and cross inheritance logic** is applied to real brain MRI images for highlighting and detecting tumor regions. The design uses no external neural network inference or BRAM/DSP blocks, maintaining both image quality and hardware efficiency.

Objectives:

- To enhance brain MRI images using inheritance-based convolution.
- To segment tumor regions using thresholding and morphological operations.
- To replace traditional convolution with hardware-efficient inheritance logic.
- To validate simulation output through post-processing in MATLAB.

Dataset used:

**Source:** <https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection>

MRI images were sourced from Kaggle’s publicly available brain tumor datasets. The dataset consists of **250 grayscale MRI images**, including:

- **Meningioma**
- **Glioma**
- **Pituitary tumors**

All images are 512×512 in size and vary in brightness, contrast, and tumor size.

The full processing pipeline includes:

### Image Preprocessing

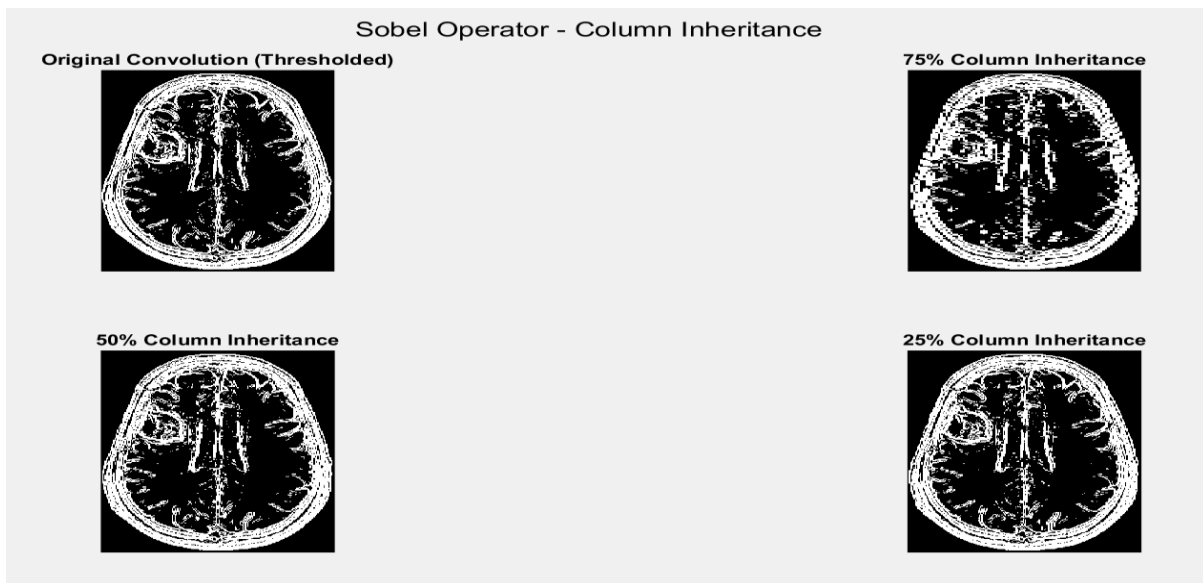
- **Conversion to Grayscale** (if RGB)
- **Contrast Enhancement** using `imadjust` in MATLAB
- **Sharpening** using a custom 3×3 kernel applied via convolution

### Inheritance-Based Convolution

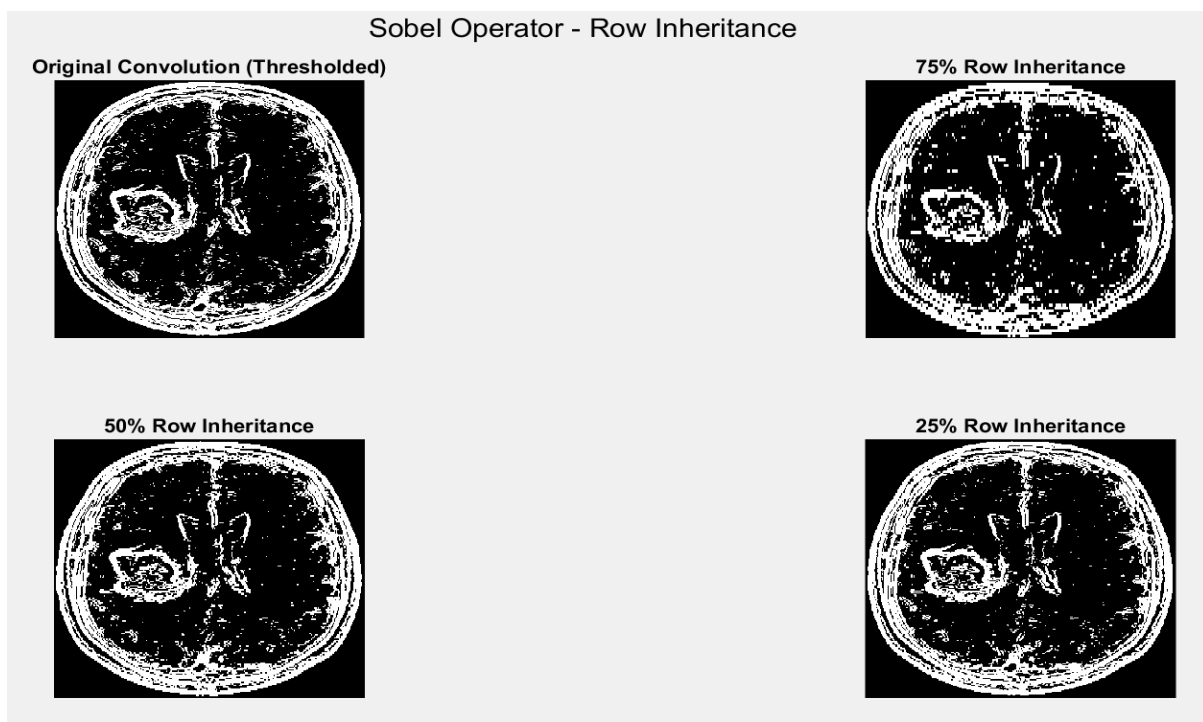
- The convolution stage is **not standard**: Instead of recalculating all 9 values for every pixel, the algorithm applies:

- 75% column inheritance
- 50% column inheritance
- 25% column inheritance
- A binary threshold ( $>4000$ ) is applied post-convolution to isolate tumor intensity regions.

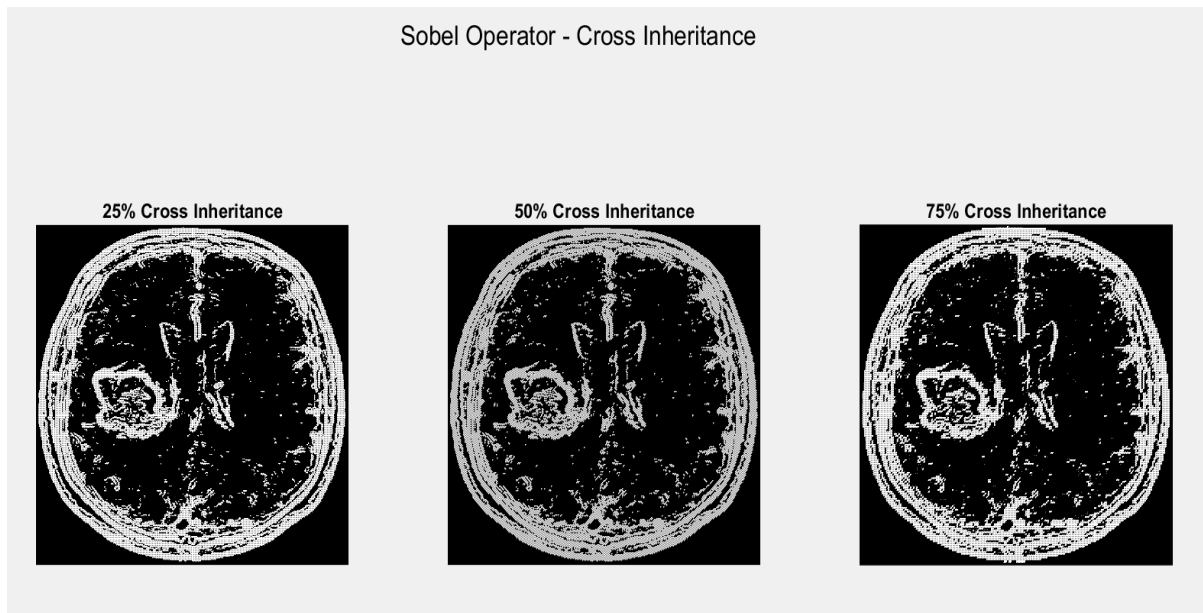
## RESULTS OF MEDICAL APPLICATION IN MATLAB:



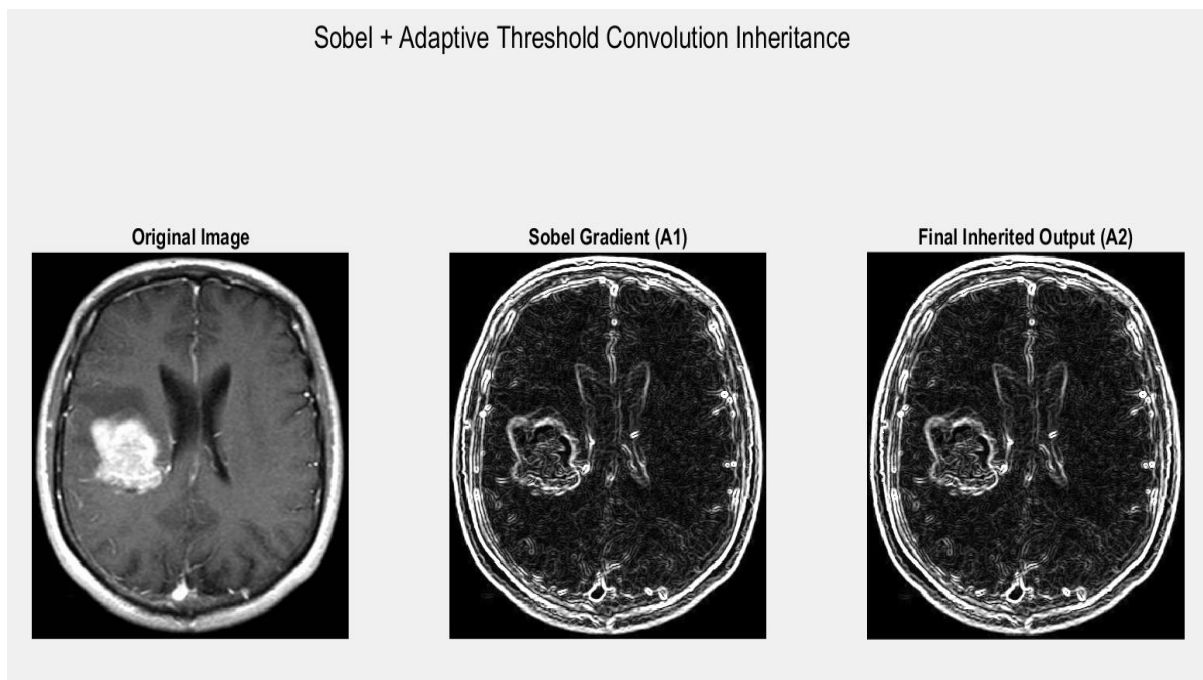
*Fig 4.12 Column Inheritance of an MRI scan*



*Fig 4.13 Row Inheritance of MRI scan*



*Fig 4.14 Crossing Inheritance of MRI scan*



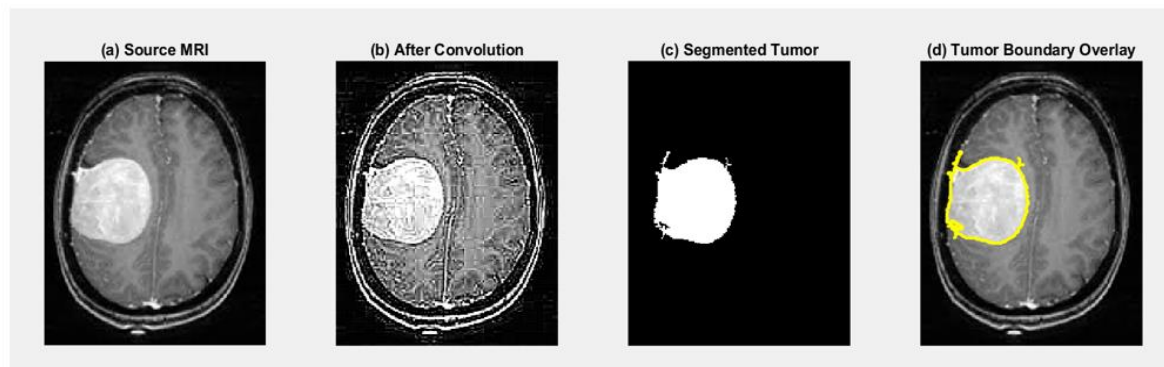
*Fig 4.15 Sobel with Adaptive Threshold*

#### **Morphological Filtering (in MATLAB)**

- **Hole filling** using `imfill`
- **Noise removal** using `bwareaopen`
- **Border clearing** using `imclearborder`
- **Largest component extraction** using `bwconncomp` and `bwboundaries`.

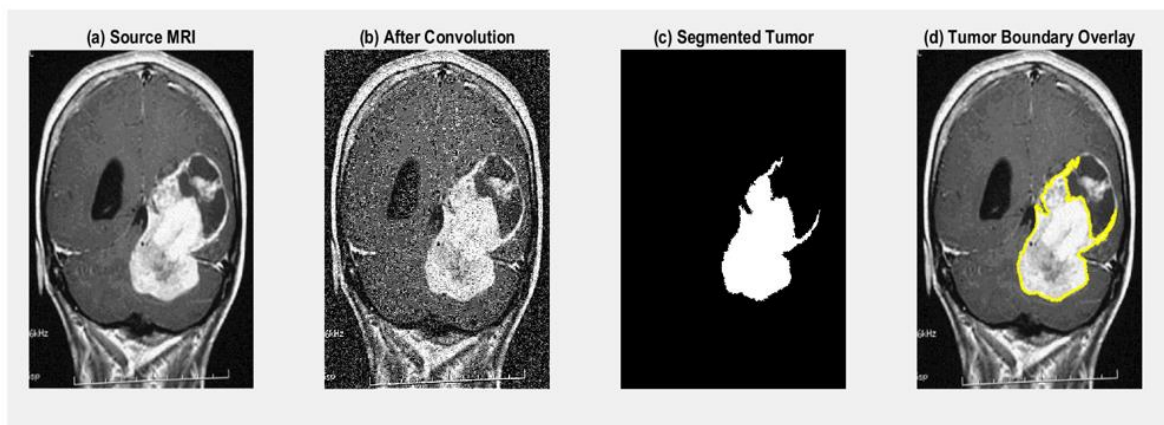


## Sample output images:



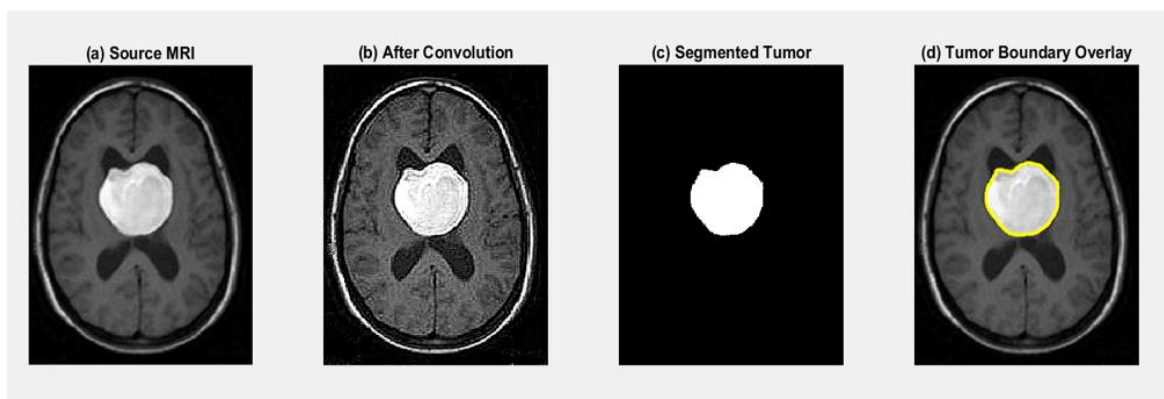
Meningioma

*Fig 4.16 Sample detection of tumor(Meningioma)*



Glioma

*Fig 4.17 Sample detection of tumor (Glioma)*



Pituitary

*Fig 4.18 Sample detection of tumor(Pituitary)*



## **CHAPTER 5**

### **DISCUSSION**

#### **5.1 PERFORMANCE VS. COMPLEXITY TRADE-OFFS**

The implementation of inheritance-based convolution techniques significantly reduced the number of computations typically required in standard 2D convolution. By selectively reusing previously computed pixel values (row-wise, column-wise, and cross-wise), the design eliminated a large portion of redundant operations. This naturally translated into reduced logic utilization and lower power consumption in hardware.

However, these computational savings came with a marginal compromise in image quality. As observed through SSIM and PSNR comparisons, higher levels of inheritance (e.g., 75%) introduced slight distortions or smoothing artifacts in the output. These effects were effectively mitigated through the Matrix Rebuild algorithm, which fused the output of all three inheritance levels into a robust final feature map.

#### **5.2 EFFECTIVENESS OF INHERITANCE-BASED LOGIC**

The use of inheritance logic proved beneficial in multiple aspects:

- It avoided full-frame buffering, which would otherwise require block RAM.
- It simplified the control logic, enabling a lightweight FSM-based streaming design.
- It achieved close approximation to traditional convolution at a fraction of the resource cost.

Each inheritance variant served a specific purpose:

- 25% preserved visual quality
- 50% maintained balance
- 75% minimized logic with acceptable degradation

Combining these levels ensured better results than any single configuration alone.

#### **5.3 VALUE OF MATRIX REBUILD AND PATCH FUSION**

The Matrix Rebuild mechanism served as a hardware-adapted fusion strategy, converting local pixel patches into higher-dimensional vectors and applying multi-kernel dot products. This approach:

- Introduced a richer feature representation while avoiding additional convolution layers
- Retained fine-grained texture and edge continuity

- Enabled image reconstruction from inherited patches with smooth output

In hardware, the logic was successfully implemented in `conv.v` using fixed-weight kernel matrices and registered patch buffers. The result was a modular, extendable unit suitable for low-power image filtering tasks.

## **5.4 MATLAB VS. VERILOG: BEHAVIOR AND OUTPUT CONSISTENCY**

Simulated Verilog outputs are showing the characteristics quite lower than MATLAB counterparts in terms of pixel distribution and visual quality. SSIM and PSNR values remained consistently high across all test cases. Slight variations in edge detail or smoothing were expected due to:

- Integer precision and rounding in hardware
- The absence of floating-point operations
- Single-cycle convolution without pipeline delay accumulation

Nonetheless, the Verilog design remained structurally faithful, validating the effectiveness of the implemented inheritance and rebuild strategies.

## **5.5 OBSERVED CHALLENGES AND IMPLEMENTATION ISSUES**

Throughout the development process, several challenges were encountered:

- Managing streaming pixel synchronization across line buffers
- Ensuring correct pixel alignment for  $3 \times 3$  patch formation
- Designing inheritance logic without increasing pipeline stalls
- Avoiding errors due to register overwriting or improper FSM resets
- Maintaining image integrity while avoiding padding artifacts at borders

Each of these issues was addressed through controlled simulation, counter validation, and careful FSM design. Extensive testing in both MATLAB and Verilog environments ensured correctness and reproducibility.

## **CHAPTER 6**

### **CONCLUSION AND FUTURE WORK**

#### **6.1 SUMMARY OF ACHIEVEMENTS**

This project presents a hardware-efficient and functionally accurate implementation of convolution-based image processing using inheritance-based logic and matrix rebuild algorithms. The system was successfully designed, simulated, and analyzed using both MATLAB and Verilog, ensuring consistency across software and hardware domains.

The core contributions of this work include:

- Design and implementation of multiple inheritance-based convolution algorithms (25%, 50%, and 75%) in MATLAB.
- Development of a novel Matrix Rebuild Algorithm, which fuses inheritance logic with dot-product-based kernel processing.
- Complete RTL development of all modules in Verilog (conv.v, imageControl.v, lineBuffer.v, etc.), ensuring efficient pixel streaming and patch processing without BRAM or DSP usage.
- Functional testbench that reconstructs output images directly from simulation.
- Verification of output quality through SSIM and PSNR metrics, showing strong alignment between MATLAB and Verilog results.
- FPGA synthesis and power analysis using Vivado, demonstrating the design's low resource utilization, low power consumption, and suitability for edge-level image processing tasks.

The results validate that inheritance logic, when carefully managed, can preserve visual quality while significantly reducing computation and hardware cost.

#### **6.2 FUTURE WORK**

Several enhancements and extensions can be considered for future exploration:

##### **1. Extension to RGB and Multi-Channel Images**

The current system processes grayscale images. Extending the pipeline to support RGB channels would allow broader application, including object recognition and color segmentation.

##### **2. Real-Time FPGA Deployment**

While the current system is simulated in Vivado, deploying it on a physical FPGA board (e.g., Artix-7 or Zynq SoC) would allow performance benchmarking under real-time constraints, including frame rate, latency, and throughput.

### 3. Adaptive Inheritance Mechanism

Incorporating data-driven or dynamic inheritance levels (instead of fixed 25/50/75%) could allow smarter trade-offs between quality and speed, especially when supported by local pixel variance or machine learning cues.

### 4. Integration with CNN Accelerators

The Matrix Rebuild concept could be embedded into lightweight CNN models as a pre-convolutional filter, enhancing feature quality while reducing memory access and multiplication requirements.

### 5. Resource-Level Optimization

Further resource optimization through pipeline balancing, register packing, and clock gating could bring down power and area even more, especially for battery-powered embedded systems.

## APPENDIX

### A. MATLAB CODES

#### 1. Basic Convolution Algorithm :

##### Column Inheritance:

###### Without image:

```
A = [1 5 8 2 3;1 5 4 9 2;7 4 3 6 8;1 4 5 6 2;1 6 4 5 4];
```

```
kernel = [0 -1 0;-1 5 -1;0 -1 0];
```

```
A1 = conv2(A, kernel);
```

```
A275p = A1; A250p = A1; A225p = A1;
```

```
for i = 1:7
```

```
for j = 1:7
```

```
% 75% Inheritance along columns
```

```
if mod(j, 4) == 0
```

```
A275p(i, j) = A1(i, j - 3);
```

```
elseif mod(j, 4) == 2
```

```
A275p(i, j) = A1(i, j - 1);
```

```
elseif mod(j, 4) == 3
```

```
A275p(i, j) = A1(i, j - 2);
```

```
end
```

```
% 50% Inheritance along columns
```

```
if mod(j, 2) == 0
```

```
A250p(i, j) = A1(i, j - 1);
```

```
end
```

```
% 25% Inheritance along columns
```

```
if mod(j, 4) == 0
```

```
A225p(i, j) = A1(i, j - 1); end end end
```

```
disp('75% Column Inheritance:');
```

```
disp(A275p);
```

```
disp('50% Column Inheritance:');
```

```
disp(A250p);
```

```
disp('25% Column Inheritance:');
```

```
disp(A225p);
```

75% Column Inheritance:

|    |    |    |    |     |     |     |
|----|----|----|----|-----|-----|-----|
| 0  | 0  | 0  | 0  | -2  | -2  | -2  |
| -1 | -1 | -1 | -1 | -10 | -10 | -10 |
| -1 | -1 | -1 | -1 | 31  | 31  | 31  |
| -7 | -7 | -7 | -7 | 4   | 4   | 4   |
| -1 | -1 | -1 | -1 | 12  | 12  | 12  |
| -1 | -1 | -1 | -1 | 11  | 11  | 11  |
| 0  | 0  | 0  | 0  | -5  | -5  | -5  |

50% Column Inheritance:

|    |    |    |    |     |     |    |
|----|----|----|----|-----|-----|----|
| 0  | 0  | -5 | -5 | -2  | -2  | 0  |
| -1 | -1 | 11 | 11 | -10 | -10 | -3 |
| -1 | -1 | 11 | 11 | 31  | 31  | -2 |
| -7 | -7 | 1  | 1  | 4   | 4   | -8 |
| -1 | -1 | 4  | 4  | 12  | 12  | -2 |
| -1 | -1 | 21 | 21 | 11  | 11  | -4 |
| 0  | 0  | -6 | -6 | -5  | -5  | 0  |

25% Column Inheritance:

|    |    |    |    |     |     |    |
|----|----|----|----|-----|-----|----|
| 0  | -1 | -5 | -5 | -2  | -3  | 0  |
| -1 | -1 | 11 | 11 | -10 | 11  | -3 |
| -1 | -8 | 11 | 11 | 31  | -10 | -2 |
| -7 | 29 | 1  | 1  | 4   | 30  | -8 |
| -1 | -7 | 4  | 4  | 12  | -8  | -2 |
| -1 | -2 | 21 | 21 | 11  | 13  | -4 |
| 0  | -1 | -6 | -6 | -5  | -4  | 0  |

###### With image:

```
img = imread('D:/rose.jpeg');
```

```
if size(img, 3) == 3
```

```
img = rgb2gray(img);
```

```
end
```

```
A = double(img);
```

```
kernel = [1 1 1;1 1 1;1 1 1];
```

```

kernel=kernel/sum(kernel(:));
A1 = conv2(A, kernel, 'full');
A2_75 = A1;
A2_50 = A1;
A2_25 = A1;
[rows, cols] = size(A1);
for i = 1:rows
for j = 1:cols
if mod(j, 4) == 0 && j - 3 > 0
A2_75(i, j) = A1(i, j - 3);
elseif mod(j, 4) == 2 && j - 1 > 0
A2_75(i, j) = A1(i, j - 1);
elseif mod(j, 4) == 3 && j - 2 > 0
A2_75(i, j) = A1(i, j - 2);
end

```

### Row Inheritance:

Without image:

```
A = [1 5 8 2 3;1 5 4 9 2;7 4 3 6 8;1 4 5 6 2;1 6 4 5 4]; kernel = [0 -1 0;-1 5 -1;0 -1 0];
```

```
A1 = conv2(A, kernel);
```

```
A275p = A1; A250p = A1; A225p = A1;
```

```
for i = 1:7
```

```
for j = 1:7
```

```
% 75% row Inheritance
```

```
if mod(i, 4) == 0
```

```
A275p(i, j) = A1(i - 3, j);
```

```
elseif mod(i, 4) == 2
```

```
A275p(i, j) = A1(i - 1, j);
```

```
elseif mod(i, 4) == 3
```

```
A275p(i, j) = A1(i - 2, j);
```

```
end
```

```
% 50% row Inheritance
```

```
if mod(i, 2) == 0
```

```
A250p(i, j) = A1(i - 1, j);
```

```
end
```

```
% 25% row Inheritance
```

```
if mod(i, 4) == 0
```

```
A225p(i, j) = A1(i - 1, j);end end end
```

```
disp('75% Inheritance:'); disp(A275p);
```

```
disp('50% Inheritance:'); disp(A250p);
```

```
disp('25% Inheritance:'); disp(A225p);
```

75% Inheritance:

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| 0  | -1 | -5 | -8 | -2 | -3 | 0  |
| 0  | -1 | -5 | -8 | -2 | -3 | 0  |
| 0  | -1 | -5 | -8 | -2 | -3 | 0  |
| 0  | -1 | -5 | -8 | -2 | -3 | 0  |
| -1 | -7 | 4  | 8  | 12 | -8 | -2 |
| -1 | -7 | 4  | 8  | 12 | -8 | -2 |
| -1 | -7 | 4  | 8  | 12 | -8 | -2 |

50% Inheritance:

|    |    |    |    |    |     |    |
|----|----|----|----|----|-----|----|
| 0  | -1 | -5 | -8 | -2 | -3  | 0  |
| 0  | -1 | -5 | -8 | -2 | -3  | 0  |
| -1 | -8 | 11 | -5 | 31 | -10 | -2 |
| -1 | -8 | 11 | -5 | 31 | -10 | -2 |
| -1 | -7 | 4  | 8  | 12 | -8  | -2 |
| -1 | -7 | 4  | 8  | 12 | -8  | -2 |
| 0  | -1 | -6 | -4 | -5 | -4  | 0  |

25% Inheritance:

|    |    |    |    |     |     |    |
|----|----|----|----|-----|-----|----|
| 0  | -1 | -5 | -8 | -2  | -3  | 0  |
| -1 | -1 | 11 | 29 | -10 | 11  | -3 |
| -1 | -8 | 11 | -5 | 31  | -10 | -2 |
| -1 | -8 | 11 | -5 | 31  | -10 | -2 |
| -1 | -7 | 4  | 8  | 12  | -8  | -2 |
| -1 | -2 | 21 | 4  | 11  | 13  | -4 |
| 0  | -1 | -6 | -4 | -5  | -4  | 0  |

With image:

```
img = imread('D:/rose.jpeg');
if size(img, 3) == 3
img = rgb2gray(img);
end
A = double(img);
kernel = [1 1 1; 1 1 1; 1 1 1];
kernel = kernel / sum(kernel(:));
A1 = conv2(A, kernel, 'full');
A2_75 = A1;
A2_50 = A1;
A2_25 = A1;
[rows, cols] = size(A1);
for i = 1:rows
for j = 1:cols
% 75% Row Inheritance
if mod(i, 4) == 0 && i - 3 > 0
A2_75(i, j) = A1(i - 3, j);
elseif mod(i, 4) == 2 && i - 1 > 0
A2_75(i, j) = A1(i - 1, j);
elseif mod(i, 4) == 3 && i - 2 > 0
A2_75(i, j) = A1(i - 2, j);
End
% 50% Row Inheritance
if mod(i, 2) == 0 && i - 1 > 0
A2_50(i, j) = A1(i - 1, j);
end
% 25% Row Inheritance
if mod(i, 4) == 0 && i - 1 > 0
A2_25(i, j) = A1(i - 1, j);
end
end
end
figure;
sgtitle("Algorithm 1- Row Inheritance");
subplot(2,2,1); imshow(uint8(A1)); title('Original Convolution');
subplot(2,2,2); imshow(uint8(A2_75)); title('75% Row Inheritance');
subplot(2,2,3); imshow(uint8(A2_50)); title('50% Row Inheritance');
subplot(2,2,4); imshow(uint8(A2_25)); title('25% Row Inheritance');
```

### **Crossing Inheritance:**

Without image:

```
clc; clear; close all;
A = [1 5 8 2 3;1 5 4 9 2;7 4 3 6 8;1 4 5 6 2;1 6 4 5 4];
kernel = [1 1 1;1 1 1;1 1 1];
kernel=kernel/sum(kernel(:));
A1 = conv2(A, kernel);
A25 = A1;
A50 = A1;
A75 = A1;
[rows, cols] = size(A1);
for i = 2:rows
    for j = 2:cols
        %% 25% CROSS INHERITANCE
        if mod(i, 2) == 0 && mod(j, 2) == 0
            A25(i,j) = (A1(i-1,j) + A1(i,j-1))/2;
        end
        %%50% CROSS INHERITANCE
        if mod(i+j,2)==0
            A50(i,j) = (A1(i-1,j) + A1(i,j-1))/2;
        end
        %% 75% CROSS INHERITANCE
        if mod(i, 2) == 0 && mod(j, 2) ~= 0
            A75(i,j) = A1(i-1,j); % inherit from top
        elseif mod(i, 2) ~= 0 && mod(j, 2) == 0
            A75(i,j) = A1(i,j-1); % inherit from left
        elseif mod(i,2)==0 && mod(j,2)==0
            A75(i,j) = (A1(i-1,j) + A1(i,j-1))/2; % average
        end
    end
end
disp('75% Inheritance:'); disp(A75);
disp('50% Inheritance:'); disp(A50);
disp('25% Inheritance:'); disp(A25);
```



75% Inheritance:

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| 0.1 | 0.7 | 1.6 | 1.7 | 1.4 | 0.6 | 0.3 |
| 0.2 | 0.4 | 1.6 | 2.2 | 1.4 | 1.8 | 0.3 |
| 1   | 1   | 4.2 | 4.2 | 5   | 5   | 1.4 |
| 1   | 1.8 | 4.2 | 4.4 | 5   | 4.2 | 1.4 |
| 1   | 1   | 3.9 | 3.9 | 4.8 | 4.8 | 1.6 |
| 0.2 | 1.4 | 3.9 | 3.6 | 4.8 | 3.2 | 1.6 |
| 0.1 | 0.1 | 1.2 | 1.2 | 1.4 | 1.4 | 0.4 |

50% Inheritance:

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| 0.1 | 0.7 | 1.6 | 1.7 | 1.4 | 0.6 | 0.3 |
| 0.2 | 0.4 | 2.7 | 2.2 | 3.1 | 1.8 | 0.6 |
| 1   | 2.6 | 2.6 | 5.1 | 4.1 | 3.3 | 1.9 |
| 1   | 1.8 | 3.8 | 4.4 | 5   | 4.2 | 1.3 |
| 1   | 2.6 | 3.2 | 4.8 | 4.9 | 3.4 | 2.4 |
| 0.2 | 1.4 | 2.3 | 3.6 | 2.9 | 3.2 | 0.7 |
| 0.1 | 0.8 | 1.6 | 1.7 | 2.3 | 1   | 0.8 |

25% Inheritance:

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| 0.1 | 0.7 | 1.6 | 1.7 | 1.4 | 0.6 | 0.3 |
| 0.2 | 0.4 | 2.7 | 2.2 | 3.1 | 1.8 | 0.6 |
| 1   | 2.6 | 4.2 | 5.1 | 5   | 3.3 | 1.4 |
| 1   | 1.8 | 3.8 | 4.4 | 5   | 4.2 | 1.3 |
| 1   | 2.6 | 3.9 | 4.8 | 4.8 | 3.4 | 1.6 |
| 0.2 | 1.4 | 2.3 | 3.6 | 2.9 | 3.2 | 0.7 |
| 0.1 | 0.8 | 1.2 | 1.7 | 1.4 | 1   | 0.4 |

With image:

```

clc; clear; close all;
img = imread('D:/lena_gray.bmp');
img = double(img);
kernel = [1 1 1;1 1 1;1 1 1];
kernel=kernel/sum(kernel(:));
A1 = conv2(img, kernel);
A25 = A1;
A50 = A1;
A75 = A1;
[rows, cols] = size(A1);
for i = 2:rows
for j = 2:cols
%% 25% CROSS INHERITANCE
if mod(i, 2) == 0 && mod(j, 2) == 0
A25(i,j) = (A1(i-1,j) + A1(i,j-1))/2;
end
%%50% CROSS INHERITANCE

```

```

if mod(i+j,2)==0
A50(i,j) = (A1(i-1,j) + A1(i,j-1))/2;
End

```

## 2. Matrix Rebuild Algorithm:

Without image:

```

clc; clear all; close all;
%% Input 3x3 Matrices
a = [1 2 0;1 1 3;0 2 2];
b = [0 2 1;0 3 2;1 1 0];
c = [1 2 1;0 1 3;3 3 2];
%% Patch Extraction Using Overlapping IDX Mapping
IDX = [1 4 2 5;4 7 5 8;2 5 3 6;5 8 6 9];
d = a(IDX)'; % Channel R
f = b(IDX)'; % Channel G
e = c(IDX)'; % Channel B
disp('Extracted Patch Matrix d :'); disp(d);
disp('Extracted Patch Matrix f :'); disp(f);
disp('Extracted Patch Matrix e :'); disp(e);
%% Input Features Matrix g (Each row is a flattened patch)
g = horzcat(d, f, e);
disp('Input Features Matrix g:'); disp(g);
%% Kernel Matrices (flattened)
k1 = [1 1; 2 2]; k2 = [1 1; 1 1]; k3 = [0 1; 1 0];
% Flatten each kernel in raster order and stack
h = k1([1 3 2 4])'; i = k2([1 3 2 4])'; j = k3([1 3 2 4])';
k = vertcat(h, i, j);
% Additional kernel set
k4 = [1 0; 0 1]; k5 = [2 1; 2 1]; k6 = [1 2; 2 0];
m = k4([1 3 2 4])'; n = k5([1 3 2 4])'; o = k6([1 3 2 4])';
p = vertcat(m, n, o);
%% Full Kernel Matrix q
q = horzcat(k, p);
disp('Kernel Matrix q:'); disp(q);
%% Matrix Multiplication Result (Simulated Convolution)
r = g * q;
disp('Initial Output Matrix r:'); disp(r);
%% Rebuild the Removed Rows by Inheriting from Row Above
u = r; % Copy of r
for i = 1:size(r, 1)

```

```

if mod(i, 3) == 2 && i > 1
u(i, :) = r(i - 1, :);
end
end
disp("Rebuilt output feature matrix:")
disp(u);

```

Input Features Matrix g:

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 1 | 1 | 0 | 2 | 0 | 3 | 1 | 2 | 0 | 1 |
| 2 | 0 | 1 | 3 | 2 | 1 | 3 | 2 | 2 | 1 | 1 | 3 |
| 1 | 1 | 0 | 2 | 0 | 3 | 1 | 1 | 0 | 1 | 3 | 3 |
| 1 | 3 | 2 | 2 | 3 | 2 | 1 | 0 | 1 | 3 | 3 | 2 |

Kernel Matrix q:

|   |   |
|---|---|
| 1 | 1 |
| 1 | 0 |
| 2 | 0 |
| 2 | 1 |
| 1 | 2 |
| 1 | 1 |
| 1 | 2 |
| 1 | 1 |
| 0 | 1 |
| 1 | 2 |
| 1 | 2 |
| 0 | 0 |

Initial Output Matrix r:

|    |    |
|----|----|
| 14 | 12 |
| 20 | 24 |
| 15 | 17 |
| 24 | 26 |

Rebuilt output feature matrix:

|    |    |
|----|----|
| 14 | 12 |
| 14 | 12 |
| 15 | 17 |
| 24 | 26 |

With image:

```
clc; clear all; close all;
img = imread('D:\brain_tumor_dataset\yes\Y11.jpg');
A = double(img);
kernel = ones(3, 3) / 9;
A1 = conv2(A, kernel, 'valid'); % 510x510
% Initialize inherited versions
A2_25 = A1;
A2_50 = A1;
A2_75 = A1;
[rows, cols] = size(A1);
% Apply inheritance logic
for i = 1:rows
    for j = 1:cols
        % 75% Inheritance
        if mod(i, 4) == 0 && i - 3 > 0
            A2_75(i, j) = A1(i - 3, j);
        elseif mod(i, 4) == 2 && i - 1 > 0
            A2_75(i, j) = A1(i - 1, j);
        elseif mod(i, 4) == 3 && i - 2 > 0
            A2_75(i, j) = A1(i - 2, j);
        end
        % 50% Inheritance
        if mod(i, 2) == 0 && i - 1 > 0
            A2_50(i, j) = A1(i - 1, j);
        end
        % 25% Inheritance
        if mod(i, 4) == 0 && i - 1 > 0
            A2_25(i, j) = A1(i - 1, j);
        end
    end
end
% Total patches =  $(510 - 2)^2 = 258064$ 
num_patches = (rows - 2) * (cols - 2);
IDX = zeros(num_patches, 9);
index_matrix = reshape(1:rows * cols, rows, cols);
k = 1;
for j = 1:(cols - 2)
    for i = 1:(rows - 2)
```

```

        block = index_matrix(i:i+2, j:j+2);
        IDX(k, :) = block(:)';
        k = k + 1;
    end
end
% Extract 3×3 patches from each matrix
patch25 = A2_25(IDX); % size: 258064 × 9
patch50 = A2_50(IDX);
patch75 = A2_75(IDX);
% Final concatenated output: 258064 × 27
G = [patch25, patch50, patch75];
disp("Final patch matrix G (258064 × 27):");
disp(G);
% Define kernels
k1 = [-1 -1 -1; -1 8 -1; -1 -1 -1];
k2 = [ 0 -1  0; -1 5 -1;  0 -1  0];
k3 = [0.1111111111 0.1111111111 0.1111111111; 0.1111111111
0.1111111111 0.1111111111; 0.1111111111 0.1111111111
0.1111111111];
col1 = [k1(:); k2(:); k3(:)];
col2 = col1(randperm(27));
col3 = col1(randperm(27));
kem = [col1, col2, col3];
disp("Kernel matrix")
disp(kem);
X= G*kem;
disp("Output feature matrix");
disp(X);
u = X; % Copy of X
for i = 1:size(X, 1)
    if mod(i, 3) == 2 && i > 1
        u(i, :) = X(i - 1, :);
    end
end
disp("Rebuilded output feature matrix:")
disp(u);
u_img = reshape(mean(u, 2), [508, 508]); % Average over 3 features
imshow(mat2gray(u_img)); title('Rebuilded output Feature Image');

```

### 3. Adaptive Threshold Algorithm:

Without image:

```
A = [1 5 8 2 3;1 5 4 9 2;7 4 3 6 8;1 4 5 6 2;1 6 4 5 4];
kernel = [-1 1 1;-2 2 2;-1 1 1];
A1 = conv2(A, kernel);
T = adaptthresh(A1, 0.2);
A2 = A1;
[rows, cols] = size(A1);
for i = 2:rows
for j = 2:cols
c = mod(A1(i,j) - A1(i-1,j), j);
if c < T(i,j)
A2(i,j) = A1(i, j-1);
else
A2(i,j) = A1(i,j);
end end end
disp("Original Matrix A:");disp(A);
disp("After Convolution (A1):");disp(A1);
disp("Adaptive Threshold Matrix (T):");disp(T);
disp("Final Output Matrix (A2):");disp(A2);
```

---

Original Matrix A:

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 5 | 8 | 2 | 3 |
| 1 | 5 | 4 | 9 | 2 |
| 7 | 4 | 3 | 6 | 8 |
| 1 | 4 | 5 | 6 | 2 |
| 1 | 6 | 4 | 5 | 4 |

After Convolution (A1):

|     |     |    |    |    |    |    |
|-----|-----|----|----|----|----|----|
| -1  | -4  | -2 | 11 | 7  | 5  | 3  |
| -3  | -12 | -2 | 22 | 25 | 21 | 8  |
| -10 | -9  | 10 | 12 | 30 | 41 | 15 |
| -16 | -1  | 18 | 5  | 22 | 47 | 20 |
| -10 | -8  | 11 | 12 | 24 | 39 | 16 |
| -3  | -13 | 6  | 13 | 19 | 26 | 10 |
| -1  | -5  | 3  | 5  | 5  | 9  | 4  |

Adaptive Threshold Matrix (T):

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |

Final Output Matrix (A2) :

|     |     |    |    |    |    |    |
|-----|-----|----|----|----|----|----|
| -1  | -4  | -2 | 11 | 7  | 5  | 3  |
| -3  | -12 | -2 | 22 | 25 | 21 | 8  |
| -10 | -9  | 10 | 12 | 30 | 41 | 15 |
| -16 | -1  | 18 | 5  | 22 | 47 | 20 |
| -10 | -8  | -8 | 11 | 12 | 39 | 16 |
| -3  | -13 | 6  | 13 | 19 | 19 | 10 |
| -1  | -5  | -5 | 3  | 5  | 5  | 9  |

With image:

```
img = imread('D:/lena_gray.jpeg');
if size(img, 3) == 3
img = rgb2gray(img);
end
A = double(img);
kernel = [1 1 1;1 1 1;1 1 1];
kernel=kernel/sum(kernel(:));
A1 = conv2(A, kernel);
T = adaptthresh(A1, 0.2);
A2 = A1; [rows, cols] = size(A1);
for i = 2:rows
for j = 2:cols
c = mod(A1(i,j) - A1(i-1,j), j);
if c < T(i,j)
A2(i,j) = A1(i, j-1);
else
A2(i,j) = A1(i,j);
endendend
figure;
subplot(1,3,1); imshow(uint8(A)); title('Original Image');
subplot(1,3,2); imshow(uint8(A1)); title('After Convolution (A1)');
subplot(1,3,3); imshow(uint8(A2)); title('Final Output (A2)');
```

## **BRAIN TUMOR DETECTION CODES:**

### **Sobel Operator Column Inheritance:**

```
clc;clear all;close all;
img = imread('D:\brain_tumor_dataset\yes\Y11.jpg');
if size(img, 3) == 3
img = rgb2gray(img);
end
A = double(img);
kernel1 = [1 0 -1; 2 0 -2; 1 0 -1];
kernel2 = [1 2 1; 0 0 0; -1 -2 -1];
conv1 = conv2(A, kernel1, 'same');
conv2 = conv2(A, kernel2, 'same');
convolved_data_int = conv1.^2 + conv2.^2;
A1 = uint8(convolved_data_int > 4000) * 255;
A2_75 = A1;
A2_50 = A1;
A2_25 = A1;
[rows, cols] = size(A1);
for i = 1:rows
for j = 1:cols
if mod(j, 4) == 0 && j - 3 > 0
A2_75(i, j) = A1(i, j - 3);
elseif mod(j, 4) == 2 && j - 1 > 0
A2_75(i, j) = A1(i, j - 1);
elseif mod(j, 4) == 3 && j - 2 > 0
A2_75(i, j) = A1(i, j - 2);
end
if mod(j, 2) == 0 && j - 1 > 0
A2_50(i, j) = A1(i, j - 1);
end
if mod(j, 4) == 0 && j - 1 > 0
A2_25(i, j) = A1(i, j - 1);
end
end
end
figure;
sgtitle("Sobel Operator - Column Inheritance");
subplot(2,2,1); imshow(A1); title('Original Convolution (Thresholded)');
subplot(2,2,2); imshow(A2_75); title('75% Column Inheritance');
```



```
subplot(2,2,3); imshow(A2_50); title('50% Column Inheritance');
subplot(2,2,4); imshow(A2_25); title('25% Column Inheritance');
```

### **MORPHOLOGICAL DETECTION CODE:**

```
img = imread('D:\brain_tumor_dataset\yes\Y1.jpg');
gray = rgb2gray(img);
% Enhance contrast
enhanced = imadjust(gray);
% Sharpening convolution
kernel = [-1 -1 -1; -1 9 -1; -1 -1 -1];
sharpened = imfilter(enhanced, kernel, 'same');
% Threshold
tumor_mask = imbinarize(sharpened, 0.72);
% Morphological cleanup
tumor_mask = imfill(tumor_mask, 'holes');
tumor_mask = bwareaopen(tumor_mask, 300);
tumor_mask = imclearborder(tumor_mask); %
CC = bwconncomp(tumor_mask);
if CC.NumObjects > 0
numPixels = cellfun(@numel, CC.PixelIdxList);
[~, idx] = max(numPixels);
tumor_mask(:) = 0; % reset
tumor_mask(CC.PixelIdxList{idx}) = 1;
end
% Get tumor boundary
boundaries = bwboundaries(tumor_mask);
figure;
subplot(1,4,1);
imshow(gray); title('(a) Source MRI');
subplot(1,4,2);
imshow(sharpened); title('(b) After Convolution');
subplot(1,4,3);
imshow(tumor_mask); title('(c) Segmented Tumor');
subplot(1,4,4);
imshow(gray); title('(d) Tumor Boundary Overlay');
hold on;
for k = 1:length(boundaries)
boundary = boundaries{k};
plot(boundary(:,2), boundary(:,1), 'y', 'LineWidth', 2);
end
```

## B. VERILOG SOURCE CODES:

### a) LineBuffer:

```
module lineBuffer(
    input  i_clk,
    input  i_rst,
    input [7:0] i_data,
    input  i_data_valid,
    output [23:0] o_data,
    input i_rd_data
);
    reg [7:0] line [511:0]; //line buffer
    reg [8:0] wrPntr;
    reg [8:0] rdPntr;
    always @(posedge i_clk)
    begin
        if(i_data_valid)
            line[wrPntr] <= i_data;
    end
    always @(posedge i_clk)
    begin
        if(i_rst)
            wrPntr <= 'd0;
        else if(i_data_valid)
            wrPntr <= wrPntr + 'd1;
    end
    assign o_data = {line[rdPntr],line[rdPntr+1],line[rdPntr+2]};

    always @(posedge i_clk)
    begin
        if(i_rst)
            rdPntr <= 'd0;
        else if(i_rd_data)
            rdPntr <= rdPntr + 'd1;
    end
endmodule
```

### b) ImageControl:

```
module imageControl(
    input          i_clk,
    input          i_rst,
    input [7:0]     i_pixel_data,
    input          i_pixel_data_valid,
    output reg [71:0] o_pixel_data,
    output          o_pixel_data_valid,
    output reg      o_intr);
```

```

reg [8:0] pixelCounter;
reg [1:0] currentWrLineBuffer;
reg [3:0] lineBuffDataValid;
reg [3:0] lineBuffRdData;
reg [1:0] currentRdLineBuffer;
wire [23:0] lb0data;
wire [23:0] lb1data;
wire [23:0] lb2data;
wire [23:0] lb3data;
reg [8:0] rdCounter;
reg rd_line_buffer;
reg [11:0] totalPixelCounter;
reg rdState;

localparam IDLE = 'b0,
          RD_BUFFER = 'b1;

assign o_pixel_data_valid = rd_line_buffer;

always @(posedge i_clk)
begin
    if(i_rst)
        totalPixelCounter <= 0;
    else
        begin
            if(i_pixel_data_valid & !rd_line_buffer)
                totalPixelCounter <= totalPixelCounter + 1;
            else if(!i_pixel_data_valid & rd_line_buffer)
                totalPixelCounter <= totalPixelCounter - 1;
        end
    end
end

always @(posedge i_clk)
begin
    if(i_rst)
        begin
            rdState <= IDLE;
            rd_line_buffer <= 1'b0;
            o_intr <= 1'b0;
        end
    else
        begin
            case(rdState)
                IDLE:begin
                    o_intr <= 1'b0;
                    if(totalPixelCounter >= 1536)

```

```

        begin
            rd_line_buffer <= 1'b1;
            rdState <= RD_BUFFER;
        end
    end
RD_BUFFER:begin
    if(rdCounter == 511)
        begin
            rdState <= IDLE;
            rd_line_buffer <= 1'b0;
            o_intr <= 1'b1;
        end
    end
endcase
end
end

always @(posedge i_clk)
begin
    if(i_rst)
        pixelCounter <= 0;
    else
        begin
            if(i_pixel_data_valid)
                pixelCounter <= pixelCounter + 1;
            end
        end
end

always @(posedge i_clk)
begin
    if(i_rst)
        currentWrLineBuffer <= 0;
    else
        begin
            if(pixelCounter == 511 & i_pixel_data_valid)
                currentWrLineBuffer <= currentWrLineBuffer+1;
            end
        end
end

always @(*)
begin
    lineBuffDataValid = 4'h0;
    lineBuffDataValid[currentWrLineBuffer] = i_pixel_data_valid;
end

```

```

always @(posedge i_clk)
begin
    if(i_rst)
        rdCounter <= 0;
    else
        begin
            if(rd_line_buffer)
                rdCounter <= rdCounter + 1;
        end
    end
end

```

```

always @(posedge i_clk)
begin
    if(i_rst)
        begin
            currentRdLineBuffer <= 0;
        end
    else
        begin
            if(rdCounter == 511 & rd_line_buffer)
                currentRdLineBuffer <= currentRdLineBuffer + 1;
        end
    end
end

```

```

always @(*)
begin
    case(currentRdLineBuffer)
        0:begin
            o_pixel_data = {lb2data,lb1data,lb0data};
        end
        1:begin
            o_pixel_data = {lb3data,lb2data,lb1data};
        end
        2:begin
            o_pixel_data = {lb0data,lb3data,lb2data};
        end
        3:begin
            o_pixel_data = {lb1data,lb0data,lb3data};
        end
    endcase
end

```

```

always @(*)
begin
    case(currentRdLineBuffer)

```

```

0:begin
    lineBuffRdData[0] = rd_line_buffer;
    lineBuffRdData[1] = rd_line_buffer;
    lineBuffRdData[2] = rd_line_buffer;
    lineBuffRdData[3] = 1'b0;
end
1:begin
    lineBuffRdData[0] = 1'b0;
    lineBuffRdData[1] = rd_line_buffer;
    lineBuffRdData[2] = rd_line_buffer;
    lineBuffRdData[3] = rd_line_buffer;
end
2:begin
    lineBuffRdData[0] = rd_line_buffer;
    lineBuffRdData[1] = 1'b0;
    lineBuffRdData[2] = rd_line_buffer;
    lineBuffRdData[3] = rd_line_buffer;
end
3:begin
    lineBuffRdData[0] = rd_line_buffer;
    lineBuffRdData[1] = rd_line_buffer;
    lineBuffRdData[2] = 1'b0;
    lineBuffRdData[3] = rd_line_buffer;
end
endcase
end

lineBuffer lB0(
    .i_clk(i_clk),
    .i_rst(i_rst),
    .i_data(i_pixel_data),
    .i_data_valid(lineBuffDataValid[0]),
    .o_data(lb0data),
    .i_rd_data(lineBuffRdData[0])
);

lineBuffer lB1(
    .i_clk(i_clk),
    .i_rst(i_rst),
    .i_data(i_pixel_data),
    .i_data_valid(lineBuffDataValid[1]),
    .o_data(lb1data),
    .i_rd_data(lineBuffRdData[1])
);

lineBuffer lB2(

```

```

.i_clk(i_clk),
.i_rst(i_rst),
.i_data(i_pixel_data),
.i_data_valid(lineBuffDataValid[2]),
.o_data(lb2data),
.i_rd_data(lineBuffRdData[2])
);

```

```

lineBuffer lB3(
.i_clk(i_clk),
.i_rst(i_rst),
.i_data(i_pixel_data),
.i_data_valid(lineBuffDataValid[3]),
.o_data(lb3data),
.i_rd_data(lineBuffRdData[3])
);

```

Endmodule

### **c)ImageProcessTop (top module):**

```

module imageProcessTop1(
input axi_clk,
input axi_reset_n,
//slave interface
input i_data_valid,
input [7:0] i_data,
output o_data_ready,
//master interface
output o_data_valid,
output [7:0] o_data,
input i_data_ready,
//interrupt
output o_intr
);
wire [71:0] pixel_data;
wire pixel_data_valid;
wire axis_prog_full;
wire [7:0] convolved_data;
wire convolved_data_valid;

assign o_data_ready = !axis_prog_full;

imageControl IC(
.i_clk(axi_clk),
.i_rst(!axi_reset_n),
.i_pixel_data(i_data),
.i_pixel_data_valid(i_data_valid),
.o_pixel_data(pixel_data),

```

```

.o_pixel_data_valid(pixel_data_valid),
.o_intr(o_intr)
);

conv conv(
.i_clk(axi_clk),
.i_pixel_data(pixel_data),
.i_pixel_data_valid(pixel_data_valid),
.o_convolved_data(convolved_data),
.o_convolved_data_valid(convolved_data_valid)
);

outputBuffer OB (
.wr_rst_busy(),      // output wire wr_rst_busy
.rd_rst_busy(),      // output wire rd_rst_busy
.s_aclk(axi_clk),     // input wire s_aclk
.s_aresetn(axi_reset_n), // input wire s_aresetn
.s_axis_tvalid(convolved_data_valid), // input wire s_axis_tvalid
.s_axis_tready(),    // output wire s_axis_tready
.s_axis_tdata(convolved_data), // input wire [7 : 0] s_axis_tdata
.m_axis_tvalid(o_data_valid), // output wire m_axis_tvalid
.m_axis_tready(i_data_ready), // input wire m_axis_tready
.m_axis_tdata(o_data), // output wire [7 : 0] m_axis_tdata
.axis_prog_full(axis_prog_full) // output wire axis_prog_full
);

```

Endmodule

#### **d)Convolution block:**

##### **i) blurred image**

```

module conv(
input    i_clk,
input [71:0] i_pixel_data,
input    i_pixel_data_valid,
output reg [7:0] o_convolved_data,
output reg  o_convolved_data_valid
);

```

```

integer i;
reg [7:0] kernel [8:0];
reg [15:0] multData[8:0];
reg [15:0] sumDataInt;
reg [15:0] sumData;
reg multDataValid;
reg sumDataValid;
reg convolved_data_valid;

```

initial



```

    begin
        for(i=0;i<9;i=i+1)
            begin
                kernel[i] = 1;
            end
        end
    end

    always @(posedge i_clk)
    begin
        for(i=0;i<9;i=i+1)
            begin
                multData[i] <= kernel[i]*i_pixel_data[i*8+:8];
            end
        multDataValid <= i_pixel_data_valid;
    end

    always @(*)
    begin
        sumDataInt = 0;
        for(i=0;i<9;i=i+1)
            begin
                sumDataInt = sumDataInt + multData[i];
            end
    end

    always @(posedge i_clk)
    begin
        sumData <= sumDataInt;
        sumDataValid <= multDataValid;
    end

    always @(posedge i_clk)
    begin
        o_convolved_data <= sumData/9;
        o_convolved_data_valid <= sumDataValid;
    end

endmodule

```

**ii) 75% Row Inheritance module: (altered to 50,25 according to algorithm)**

```

module conv(row
    input    i_clk,
    input [71:0] i_pixel_data,
    input    i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg    o_convolved_data_valid
);

```

```

integer i;
reg [7:0] kernel [8:0];
reg [15:0] multData[8:0];
reg [15:0] sumDataInt;
reg [15:0] sumData;
reg multDataValid;
reg sumDataValid;

reg [9:0] row_counter = 0;
reg [9:0] col_counter = 0;
reg [18:0] current_index;
reg [18:0] prev_index_1, prev_index_2, prev_index_3;
reg [7:0] output_memory [0:262143];
reg [7:0] inherited_pixel;
reg [7:0] final_pixel;
reg write_enable;

initial begin
    for (i = 0; i < 9; i = i + 1)
        kernel[i] = 1;
end

always @(posedge i_clk) begin
    for (i = 0; i < 9; i = i + 1)
        multData[i] <= kernel[i] * i_pixel_data[i*8+: 8];
    multDataValid <= i_pixel_data_valid;
end

always @(*) begin
    sumDataInt = 0;
    for (i = 0; i < 9; i = i + 1)
        sumDataInt = sumDataInt + multData[i];
end

always @(posedge i_clk) begin
    sumData <= sumDataInt;
    sumDataValid <= multDataValid;
end

always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        if (col_counter == 511) begin
            col_counter <= 0;
            if (row_counter == 511)
                row_counter <= 0;
        else

```

```

        row_counter <= row_counter + 1;
    end else begin
        col_counter <= col_counter + 1;
    end
end
end

always @(posedge i_clk) begin
    if (sumDataValid) begin

        current_index = row_counter * 512 + col_counter;
        prev_index_1 = (row_counter > 0) ? ((row_counter - 1) * 512 + col_counter) : 0;
        prev_index_2 = (row_counter > 1) ? ((row_counter - 2) * 512 + col_counter) : 0;
        prev_index_3 = (row_counter > 2) ? ((row_counter - 3) * 512 + col_counter) : 0;

        inherited_pixel = sumData / 9;

        if ((row_counter % 4 == 0) && row_counter >= 3)
            final_pixel = output_memory[prev_index_3];
        else if ((row_counter % 4 == 2) && row_counter >= 1)
            final_pixel = output_memory[prev_index_1];
        else if ((row_counter % 4 == 3) && row_counter >= 2)
            final_pixel = output_memory[prev_index_2];
        else
            final_pixel = inherited_pixel;

        o_convolved_data <= final_pixel;
        o_convolved_data_valid <= 1;

        output_memory[current_index] <= final_pixel;
    end else begin
        o_convolved_data_valid <= 0;
    end
end
endmodule

```

### iii) 75% Column Inheritance (altered to 50,25% according to algorithm)

```

module conv(column
    input    i_clk,
    input [71:0] i_pixel_data,
    input    i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg    o_convolved_data_valid
);

```

```

integer i;
reg [7:0] kernel [8:0];
reg [15:0] multData[8:0];
reg [15:0] sumDataInt;
reg [15:0] sumData;
reg multDataValid;
reg sumDataValid;

reg [9:0] row_counter = 0;
reg [9:0] col_counter = 0;
reg [18:0] current_index;
reg [18:0] prev_col_1, prev_col_2, prev_col_3;
reg [7:0] output_memory [0:262143];
reg [7:0] inherited_pixel;
reg [7:0] final_pixel;

initial begin
    for (i = 0; i < 9; i = i + 1)
        kernel[i] = 1;
end

// Multiply kernel with input pixels
always @(posedge i_clk) begin
    for (i = 0; i < 9; i = i + 1)
        multData[i] <= kernel[i] * i_pixel_data[i*8 +: 8];
    multDataValid <= i_pixel_data_valid;
end

// Sum of all multiplied data
always @(*) begin
    sumDataInt = 0;
    for (i = 0; i < 9; i = i + 1)
        sumDataInt = sumDataInt + multData[i];
end

always @(posedge i_clk) begin
    sumData <= sumDataInt;
    sumDataValid <= multDataValid;
end

// Update row and column counters
always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        if (col_counter == 511) begin
            col_counter <= 0;
            if (row_counter == 511)

```

```

        row_counter <= 0;
    else
        row_counter <= row_counter + 1;
    end else begin
        col_counter <= col_counter + 1;
    end
end
end
end

// Column inheritance logic
always @(posedge i_clk) begin
    if (sumDataValid) begin
        current_index = row_counter * 512 + col_counter;
        prev_col_1 = (col_counter > 0) ? (row_counter * 512 + (col_counter - 1)) : 0;
        prev_col_2 = (col_counter > 1) ? (row_counter * 512 + (col_counter - 2)) : 0;
        prev_col_3 = (col_counter > 2) ? (row_counter * 512 + (col_counter - 3)) : 0;

        inherited_pixel = sumData / 9;

        // Inherit from previous columns based on column_counter % 4
        if ((col_counter % 4 == 0) && col_counter >= 3)
            final_pixel = output_memory[prev_col_3];
        else if ((col_counter % 4 == 2) && col_counter >= 1)
            final_pixel = output_memory[prev_col_1];
        else if ((col_counter % 4 == 3) && col_counter >= 2)
            final_pixel = output_memory[prev_col_2];
        else
            final_pixel = inherited_pixel;

        o_convolved_data <= final_pixel;
        o_convolved_data_valid <= 1;

        output_memory[current_index] <= final_pixel;
    end else begin
        o_convolved_data_valid <= 0;
    end
end
end endmodule

```

#### iv) Adaptive Threshold Convolution:

```

module conv_adaptive_threshold (
    input    i_clk,
    input [71:0] i_pixel_data,
    input    i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg    o_convolved_data_valid

```

```

);
integer i;
reg [7:0] kernel [0:8];
reg [15:0] multData [0:8];
reg [15:0] sumData;
reg [7:0] conv_pixel;

reg [9:0] row_counter = 0;
reg [9:0] col_counter = 0;
reg [18:0] current_index;

// Memory to hold intermediate and final output
reg [7:0] output_memory [0:262143]; // For 512x512 image
reg [7:0] previous_row_pixels [0:511];

// Adaptive logic
reg [7:0] pixel_ij, pixel_ilj, pixel_ijl;
reg [7:0] final_pixel;
parameter THRESHOLD = 20;

initial begin
    for (i = 0; i < 9; i = i + 1)
        kernel[i] = 1;
end

always @(posedge i_clk) begin
    for (i = 0; i < 9; i = i + 1)
        multData[i] <= kernel[i] * i_pixel_data[i*8+: 8];
end

always @(*) begin
    sumData = 0;
    for (i = 0; i < 9; i = i + 1)
        sumData = sumData + multData[i];
    conv_pixel = sumData / 9; // Box blur average
end

always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        if (col_counter == 511) begin
            col_counter <= 0;
            if (row_counter == 511)
                row_counter <= 0;
        else
            row_counter <= row_counter + 1;
        end else begin
            col_counter <= col_counter + 1;
        end
    end
end

```

```

        end
    end
end
always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        current_index = row_counter * 512 + col_counter;

        pixel_ij = conv_pixel;
        pixel_ij1 = (row_counter > 0) ? previous_row_pixels[col_counter] : conv_pixel;
        pixel_ij1 = (col_counter > 0) ? output_memory[current_index - 1] : conv_pixel;
        if (((pixel_ij - pixel_ij1) % (col_counter + 1)) < THRESHOLD)
            final_pixel = pixel_ij1; // Inherit from previous column
        else
            final_pixel = pixel_ij; // Use current convolved value
        o_convolved_data <= final_pixel;
        o_convolved_data_valid <= 1;

        output_memory[current_index] <= final_pixel;
        previous_row_pixels[col_counter] <= conv_pixel;
    end else begin
        o_convolved_data_valid <= 0;
    end end endmodule

```

#### v) 75% Cross inheritance

```

module conv (
    input      i_clk,
    input      i_rst,
    input  [71:0] i_pixel_data,
    input      i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg      o_convolved_data_valid
);

integer i;

// 3x3 kernel weights (all ones for averaging)
reg [7:0] kernel[0:8];
initial begin
    for (i = 0; i < 9; i = i + 1)
        kernel[i] = 8'd1;
end

// Intermediate variables
reg [15:0] multData[0:8];
reg [15:0] sumDataInt;

```

```

reg [15:0] sumData;
reg      multDataValid, sumDataValid;

// Inheritance state
reg [7:0] row_buffer[0:511]; // Buffer to store previous row values (for top inheritance)
reg [7:0] left_pixel;        // Previous pixel value in current row (for left inheritance)

// Control counters
reg [9:0] row_counter = 0;
reg [9:0] col_counter = 0;

// Final inherited pixel logic
reg [7:0] conv_result;
reg [7:0] top_pixel;
reg [7:0] left_pixel_curr;

// Step 1: Multiply kernel with pixel data
always @(posedge i_clk) begin
    for (i = 0; i < 9; i = i + 1)
        multData[i] <= kernel[i] * i_pixel_data[i*8+: 8];
    multDataValid <= i_pixel_data_valid;
end

// Step 2: Sum the multiplied results (combinational)
always @(*) begin
    sumDataInt = 0;
    for (i = 0; i < 9; i = i + 1)
        sumDataInt = sumDataInt + multData[i];
end

// Step 3: Store the final sum on clock edge
always @(posedge i_clk) begin
    sumData <= sumDataInt;
    sumDataValid <= multDataValid;
end

// Step 4: Update row and column counters
always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        if (col_counter == 511) begin
            col_counter <= 0;
            if (row_counter == 511)
                row_counter <= 0;
            else
                row_counter <= row_counter + 1;
        end else begin

```



```

        col_counter <= col_counter + 1;
    end
end
end

// Step 5: Final output with 75% crossing inheritance logic
always @(posedge i_clk) begin
    if (i_rst) begin
        o_convolved_data    <= 0;
        o_convolved_data_valid <= 0;
        left_pixel          <= 0;
    end else if (sumDataValid) begin
        conv_result    <= sumData / 9;
        top_pixel      <= row_buffer[col_counter];
        left_pixel_curr <= left_pixel;

        // 75% CROSSING INHERITANCE LOGIC
        if ((row_counter % 2 == 0) && (col_counter % 2 == 1))
            o_convolved_data <= top_pixel; // Inherit from top
        else if ((row_counter % 2 == 1) && (col_counter % 2 == 0))
            o_convolved_data <= left_pixel_curr; // Inherit from left
        else if ((row_counter % 2 == 0) && (col_counter % 2 == 0))
            o_convolved_data <= (top_pixel + left_pixel_curr) >> 1; // Average
        else
            o_convolved_data <= conv_result; // Use convolved result

        o_convolved_data_valid <= 1;

        // Update buffers
        row_buffer[col_counter] <= conv_result;
        left_pixel          <= conv_result;
    end else begin
        o_convolved_data_valid <= 0;
    end
end
endmodule

```

#### vi) 50% Cross Inheritance:

```

module conv (
    input      i_clk,
    input      i_rst,
    input  [71:0] i_pixel_data,
    input      i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg      o_convolved_data_valid

```

```

);

integer i;

reg [7:0] kernel[0:8];
initial begin
    for (i = 0; i < 9; i = i + 1)
        kernel[i] = 8'd1;
end

reg [15:0] multData[0:8];
reg [15:0] sumDataInt;
reg [15:0] sumData;
reg      multDataValid, sumDataValid;

reg [7:0] row_buffer[0:511];
reg [7:0] left_pixel;

reg [9:0] row_counter = 0;
reg [9:0] col_counter = 0;

reg [7:0] conv_result;
reg [7:0] top_pixel;
reg [7:0] left_pixel_curr;

always @(posedge i_clk) begin
    for (i = 0; i < 9; i = i + 1)
        multData[i] <= kernel[i] * i_pixel_data[i*8+: 8];
    multDataValid <= i_pixel_data_valid;
end

always @(*) begin
    sumDataInt = 0;
    for (i = 0; i < 9; i = i + 1)
        sumDataInt = sumDataInt + multData[i];
end

always @(posedge i_clk) begin
    sumData <= sumDataInt;
    sumDataValid <= multDataValid;
end

always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        if (col_counter == 511) begin
            col_counter <= 0;

```

```

        if (row_counter == 511)
            row_counter <= 0;
        else
            row_counter <= row_counter + 1;
        end else begin
            col_counter <= col_counter + 1;
        end
    end
end

always @(posedge i_clk) begin
    if (i_rst) begin
        o_convolved_data <= 0;
        o_convolved_data_valid <= 0;
        left_pixel <= 0;
    end else if (sumDataValid) begin
        conv_result <= sumData / 9;
        top_pixel <= row_buffer[col_counter];
        left_pixel_curr <= left_pixel;

        if (((row_counter + col_counter) % 2) == 0)
            o_convolved_data <= (top_pixel + left_pixel_curr) >> 1;
        else
            o_convolved_data <= conv_result;

        o_convolved_data_valid <= 1;
        row_buffer[col_counter] <= conv_result;
        left_pixel <= conv_result;
    end else begin
        o_convolved_data_valid <= 0;
    end
end

endmodule

```

**vii) 25% Cross Inheritance:**

```

module conv (
    input      i_clk,
    input      i_rst,
    input  [71:0] i_pixel_data,
    input      i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg      o_convolved_data_valid
);

integer i;

```

```

reg [7:0] kernel[0:8];
initial begin
    for (i = 0; i < 9; i = i + 1)
        kernel[i] = 8'd1;
end

reg [15:0] multData[0:8];
reg [15:0] sumDataInt;
reg [15:0] sumData;
reg      multDataValid, sumDataValid;

reg [7:0] row_buffer[0:511];
reg [7:0] left_pixel;

reg [9:0] row_counter = 0;
reg [9:0] col_counter = 0;

reg [7:0] conv_result;
reg [7:0] top_pixel;
reg [7:0] left_pixel_curr;

always @(posedge i_clk) begin
    for (i = 0; i < 9; i = i + 1)
        multData[i] <= kernel[i] * i_pixel_data[i*8+: 8];
    multDataValid <= i_pixel_data_valid;
end

always @(*) begin
    sumDataInt = 0;
    for (i = 0; i < 9; i = i + 1)
        sumDataInt = sumDataInt + multData[i];
end

always @(posedge i_clk) begin
    sumData <= sumDataInt;
    sumDataValid <= multDataValid;
end

always @(posedge i_clk) begin
    if (i_pixel_data_valid) begin
        if (col_counter == 511) begin
            col_counter <= 0;
            if (row_counter == 511)
                row_counter <= 0;
        else
            row_counter <= row_counter + 1;
    end
end

```

```

        end else begin
            col_counter <= col_counter + 1;
        end
    end
end

always @(posedge i_clk) begin
    if (i_rst) begin
        o_convolved_data    <= 0;
        o_convolved_data_valid <= 0;
        left_pixel          <= 0;
    end else if (sumDataValid) begin
        conv_result    <= sumData / 9;
        top_pixel      <= row_buffer[col_counter];
        left_pixel_curr <= left_pixel;

        if ((row_counter % 2 == 0) && (col_counter % 2 == 0))
            o_convolved_data <= (top_pixel + left_pixel_curr) >> 1;
        else
            o_convolved_data <= conv_result;

        o_convolved_data_valid <= 1;
        row_buffer[col_counter] <= conv_result;
        left_pixel <= conv_result;
    end else begin
        o_convolved_data_valid <= 0;
    end
end

endmodule

```

### **viii) Matrix Rebuild Algorithm (simplified form)**

```

module conv (
    input      i_clk,
    input      i_rst,
    input  [71:0] i_pixel_data,
    input      i_pixel_data_valid,
    output reg [7:0] o_convolved_data,
    output reg      o_convolved_data_valid
);

    reg [7:0] p [0:8]; // 3x3 pixels
    real G[0:26];      // 27-element feature vector
    real F[0:2];       // Output features

```

```

real mean_out;

// Kernels
real k1[0:8]; // Edge
real k2[0:8]; // Sharpen
real k3[0:8]; // Blur

integer i;

initial begin
    // Edge detection kernel
    k1[0]=-1; k1[1]=-1; k1[2]=-1;
    k1[3]=-1; k1[4]= 8; k1[5]=-1;
    k1[6]=-1; k1[7]=-1; k1[8]=-1;

    // Sharpen kernel
    k2[0]= 0; k2[1]=-1; k2[2]= 0;
    k2[3]=-1; k2[4]= 5; k2[5]=-1;
    k2[6]= 0; k2[7]=-1; k2[8]= 0;

    // Blur kernel (mean)
    for (i = 0; i < 9; i = i + 1)
        k3[i] = 1.0 / 9.0;
end

always @(posedge i_clk) begin
    if (i_rst) begin
        o_convolved_data <= 0;
        o_convolved_data_valid <= 0;
    end else if (i_pixel_data_valid) begin
        // Extract pixels
        for (i = 0; i < 9; i = i + 1)
            p[i] = i_pixel_data[i*8+: 8];

        // Inheritance simulation
        for (i = 0; i < 9; i = i + 1) begin
            G[i] = p[i]; // 25%
            G[i+9] = (i > 0) ? p[i-1] : p[i]; // 50%
            G[i+18] = (i > 1) ? p[i-2] : p[i]; // 75%
        end

        // Feature fusion
        F[0] = 0; F[1] = 0; F[2] = 0;
        for (i = 0; i < 9; i = i + 1) begin
            F[0] = F[0] + G[i] * k1[i];
            F[1] = F[1] + G[i+9] * k2[i];

```

```

        F[2] = F[2] + G[i+18] * k3[i];
    end

    mean_out = (F[0] + F[1] + F[2]) / 3.0;

    // Clamp and assign output
    if (mean_out > 255)
        o_convolved_data <= 8'd255;
    else if (mean_out < 0)
        o_convolved_data <= 8'd0;
    else
        o_convolved_data <= mean_out;

        o_convolved_data_valid <= 1'b1;
    end else begin
        o_convolved_data_valid <= 1'b0;
    end
end
endmodule

```

### **TestBench Code:**

```

`define headerSize 1080
`define imageSize 512*512

module tb(

);

    reg clk;
    reg reset;
    reg [7:0] imgData;
    integer file,file1,i;
    reg imgDataValid;
    integer sentSize;
    wire intr;
    wire [7:0] outData;
    wire outDataValid;
    integer receivedData=0;

    initial
    begin
        clk = 1'b0;
        forever
        begin
            #5 clk = ~clk;
        end
    end

```

```

end

initial
begin
    reset = 0;
    sentSize = 0;
    imgDataValid = 0;
    #100;
    reset = 1;
    #100;
    file = $fopen("lena_gray.bmp","rb");
    file1 = $fopen("blurred_lena.bmp","wb");
    for(i=0;i<`headerSize;i=i+1)
    begin
        $fscanf(file,"%c",imgData);
        $fwrite(file1,"%c",imgData);
    end

    for(i=0;i<4*512;i=i+1)
    begin
        @(posedge clk);
        $fscanf(file,"%c",imgData);
        imgDataValid <= 1'b1;
    end
    sentSize = 4*512;
    @(posedge clk);
    imgDataValid <= 1'b0;
    while(sentSize < `imageSize)
    begin
        @(posedge intr);
        for(i=0;i<512;i=i+1)
        begin
            @(posedge clk);
            $fscanf(file,"%c",imgData);
            imgDataValid <= 1'b1;
        end
        @(posedge clk);
        imgDataValid <= 1'b0;
        sentSize = sentSize+512;
    end
    @(posedge clk);
    imgDataValid <= 1'b0;
    @(posedge intr);
    for(i=0;i<512;i=i+1)
    begin
        @(posedge clk);
        imgData <= 0;
        imgDataValid <= 1'b1;
    end
    @(posedge clk);

```



```

imgDataValid <= 1'b0;
@(posedge intr);
for(i=0;i<512;i=i+1)
begin
    @(posedge clk);
    imgData <= 0;
    imgDataValid <= 1'b1;
end
@(posedge clk);
imgDataValid <= 1'b0;
$fclose(file);
end

always @(posedge clk)
begin
    if(outDataValid)
    begin
        $fwrite(file1,"%c",outData);
        receivedData = receivedData+1;
    end
    if(receivedData == `imageSize)
    begin
        $fclose(file1);
        $stop;
    end
end

imageProcessTop1 dut(
    .axi_clk(clk),
    .axi_reset_n(reset),
    //slave interface
    .i_data_valid(imgDataValid),
    .i_data(imgData),
    .o_data_ready(),
    //master interface
    .o_data_valid(outDataValid),
    .o_data(outData),
    .i_data_ready(1'b1),
    //interrupt
    .o_intr(intr)
);

endmodule

```

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